Representations of geometric shapes have syntactic structure

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Barbu Revencu, Maxence Pajot, and Stanislas Dehaene

NeuroSpin Cognitive Neuroimaging Unit, CEA, INSERM, Université Paris-Saclay

Author Note

Barbu Revencu https://orcid.org/0000-0001-8701-9123

Maxence Pajot https://orcid.org/0009-0001-3331-9128

Stanislas Dehaene https://orcid.org/0000-0002-7418-8275

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Correspondence concerning this article should be addressed to Barbu Revencu, NeuroSpin, Centre d'études de Saclay, Bâtiment 145, 91191 Gif-sur-Yvette, France. Email: barbu.revencu@college-de-france.fr.

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Abstract

Recent research suggests that humans use language-like mental representations for many classes of stimuli, from auditory sequences to visual shapes. However, evidence has been largely indirect, relying on stimulus complexity and compression as a proxy for internal representation. In this study, we probe the structure of geometric shape representations more directly. Instead of complexity measures, we rely on constituency tests similar to those used in linguistics and probe whether geometric shapes exhibit structural ambiguity, constituent subparts, and syntactic movement. Across three preregistered experiments ($n = 136$) using a rigorously controlled set of stimuli, we find robust evidence for tree structure in human adults' shape representations. First, the same shape can receive different structural representations depending on how a preceding animation organizes it. Second, subparts of shapes are easier to detect when they belong to the same subtree than when they span different subtrees. Third, shape fragments are easier to reconfigure the higher in the tree they are split. These results support the hypothesis that a geometric shape is internally represented as a hierarchical tree of constituents nested inside each other. Thus, while geometric shapes differ in many ways from natural-language sentences, they are nevertheless encoded by humans in a language-like representational format.

Keywords: geometric shapes, language of thought, syntax, tree structure, constituency tests

Public significance statement

A hallmark of human language is the ability to create countless expressions from a limited set of building blocks. Our study shows that the same capacity underlies how adults perceive geometric shapes. Despite obvious differences—language is auditory and unfolds over time, shapes are visual and appear all-at-once—both are mentally represented using similar principles: structured hierarchies built out of simpler parts. These findings suggest that the combinatorial structures often associated with language reflect a more general feature of human cognition.

Introduction

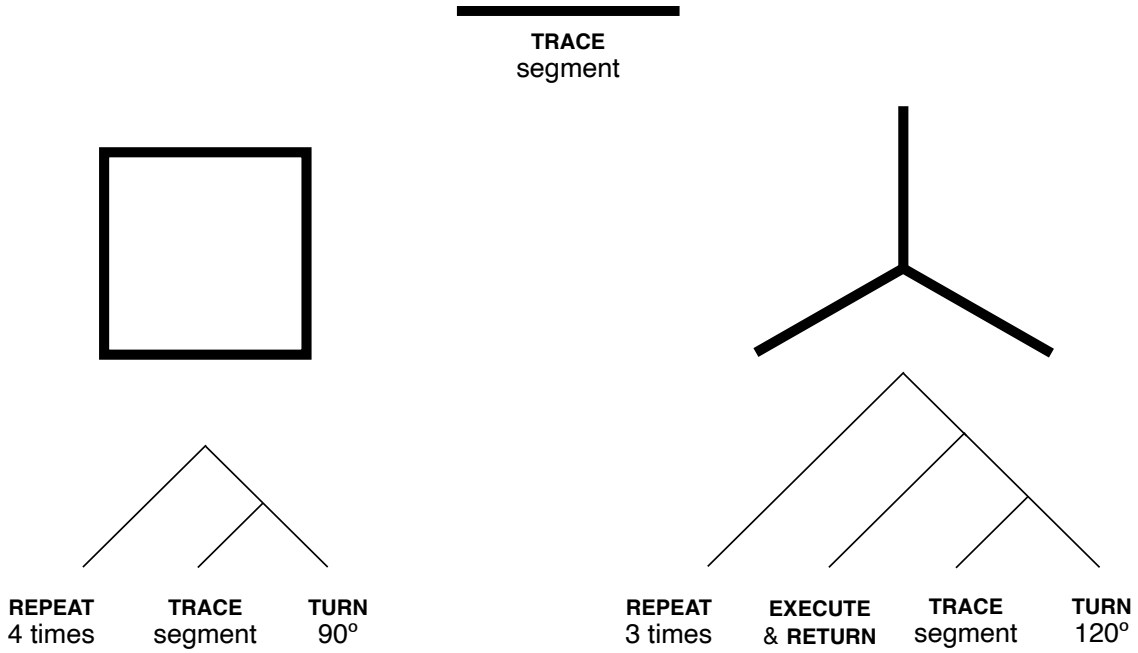
According to the Language of Thought (LoT) hypothesis (Fodor, 1975; Piantadosi et al., 2016), a subset of mental representations have language-like structure. These representations draw on a finite lexicon of primitives (e.g., DOG, CHILD, FRIGHTEN) and are built according to a finite set of rules—a grammar—specifying how the primitives can be legally combined (e.g., DOG_1 $\text{FRIGHTEN}_{\text{ed}}$ CHILD_1). The LoT hypothesis helps explain two central properties of human thought and language (Fodor & Pylyshyn, 1988): their *productivity* (the ability to generate an unbounded set of representations) and their *systematicity* (the ability to entertain DOG_1 $\text{FRIGHTEN}_{\text{ed}}$ CHILD_1 implies the ability to entertain CHILD_1 $\text{FRIGHTEN}_{\text{ed}}$ DOG_1).

Recent evidence suggests that LoT representations extend well beyond natural language. To investigate LoT representations, researchers construct formal languages, generate (nonlinguistic) stimuli based on the languages' primitives and rules, and test whether participants use these representations to encode the stimuli. Across many domains, from Boolean concepts to auditory and visuospatial sequences, human response times, error rates, and perceived complexity in behavioral tasks (Feldman, 2000), as well as neural activity in fMRI and MEG paradigms (Al Roumi et al., 2023; Wang et al., 2019), align with stimulus complexity as defined in these formal languages. This occurs because the LoT description affords the compression of complex stimuli (e.g., $ABABABAB$) into simpler expressions (repeat AB 4 times), which are easier to remember or detect, relative to stimuli devoid of such compressible regularities. This widespread pattern points to the possibility of not one but *multiple* LoTs, each governed by its own set of primitives and syntactic rules (Dehaene et al., 2022).

Building on foundational work in Gestalt psychology (e.g., Wagemans et al., 2012; Wertheimer, 1938), internal languages have also been proposed for the encoding of visual objects, sequences and scenes (Amalric et al., 2017; Cavanagh, 2021; Leeuwenberg, 1971; Leeuwenberg & van der Helm, 2013; Mills et al., 2023). In a recent iteration of this family of hypotheses, Sablé-Meyer et al. (2022) put forth a language of thought for geometry that can generate many of the shapes that have been attested cross-culturally. This language consists of a small handful of primitives (Figure 1), among which TRACE (draw a segment), TURN (change the drawing angle), REPEAT (take a set of drawing instructions as input and repeat them n times), and SUBPROGRAM (execute a set of drawing instructions, then return to the original position). Sablé-Meyer et al. (2022) tested the hypothesized geometry LoT using match-to-sample tasks: Adults were exposed to shapes that they had to remember, then select them from an array of distractors after a short delay. As in LoT work for other stimuli classes, participants' encoding times, choice times, and error rates covaried, on a shape-by-shape basis, with the complexity predicted by the LoT, i.e. the number of primitives in the most compact description of a shape, also known as "minimum description length" (MDL). Similar

evidence has been obtained with handwritten characters (Lake et al., 2015) and spatial sequences (Amalric et al., 2017; Mills et al., 2023).

Figure 1
Geometry LoT overview



Note. The Language of Thought for geometry (Sablé-Meyer et al., 2022) illustrated on three shapes. Previous work has focused on the compression afforded by the primitives of the language (bottom row). The focus on this paper is on its syntactic structure (middle row).

While evidence of correlations between the psychological difficulty of encoding/retrieving a given shape and its MDL is revealing about its internal representation, it is also highly indirect. First, inferring the representational structure from complexity measures relies on a mere correlation between a behavioral or neural parameter and a numerical index of complexity. While Bayesian model selection can be used to determine which LoT, among various possible implementations, provides the best fit to the data (Piantadosi et al., 2016; Sablé-Meyer et al., 2022), this approach provides only indirect evidence in favor of the hypothesized mental syntactic structures.

Second, using compression as a proxy for LoT representations conflates two computational issues that are partly independent: representational format (**Q**: What type of representation is involved? **A**: LoT) and selection of the internal representation (**Q**: If multiple representations are available for the same stimulus, on which basis is one of them preferred over the others? **A**: MDL). While the defining feature of any LoT is its formal structure, which presupposes the coexistence of a lexicon of primitive elements and a grammar of rules for their combination (Fodor, 1975; Quilty-Dunn et al., 2023), the evidence supporting

LoT representations mainly rests on the secondary question of efficient compression. For instance, in the shape language from Sablé-Meyer et al. (2022) illustrated in Figure 1, the optimal program for a square is REPEAT[REPEAT(TRACE, TURN)] (complexity 4, because it uses 4 primitives). A suboptimal square program such as REPEAT(TRACE, TURN, TRACE, TURN) is less efficient (complexity 5), but it counts as an LoT representation all the same, because it is generated from the same lexicon and grammar.

This reliance on complexity measures in LoT research seems especially roundabout when considering research in linguistics, where understanding how syntax works is a primary goal. Linguists have long come up with a variety of criteria for probing syntactic structures that do not presuppose efficiency (Chomsky, 1957; Haegeman, 2006). Hence, the present study aims to bridge the gap between natural language and visual LoT research by replacing complexity measures with constituency tests inspired from linguistics. Just as such tests reveal that sentences are processed as nested tree structures rather than linear word strings, we designed our experiments as visual analogs that probe for tree structure in geometric shape representations. Experiment 1 builds on structural ambiguity to show that multiple representations can be assigned to the same shape. Experiment 2 builds on clausal segmentation to show that shape subparts are easier to recover when they belong to the same subtree than when they span multiple subtrees. Finally, Experiment 3 builds on syntactic movement to show that the deeper in the tree a shape is split, the more time it takes participants to put it back together. The three experiments together strengthen the evidence for hierarchically structured representations of visual stimuli (Biederman, 1987; Cavanagh, 2021; Palmer, 1977) and for the shape LoT implementation in Sablé-Meyer et al. (2022).

Our research also provided the opportunity to address the secondary goal of testing how easily these results can be captured by state-of-the-art deep neural networks, whose capacity to model human cognition remains hotly debated (Jacob et al., 2021; Tuli et al., 2021). Recently, Campbell et al. (2024) reported that such networks could partly mimic human geometric shape complexity effects (Sablé-Meyer et al., 2021; Sablé-Meyer et al., 2022) and argued that it was therefore unnecessary to posit a LoT in humans. However, not only is this conclusion unsound, since neural networks might implement the functional equivalent of an LoT in their internal computations (Hornik et al., 1989; Quilty-Dunn et al., 2023), but, as we will see, these networks did not show any evidence of syntactic structure when tested with the stimuli of Experiments 1–3. Thus, we conclude that state-of-the-art neural networks deviate in systematic ways from human cognition and that constituency tests provide a more stringent test of the LoT hypothesis than complexity measures, where humans and neural networks partly align.

Experiment 1: Structural Ambiguity

Many natural-language sentences have the same surface form but different meanings depending on their syntactic parse: *I saw [the man with binoculars]* versus *I saw [the man] [with binoculars].* Such structural ambiguity has long been considered evidence for nested syntactic structures in natural language (Chomsky, 1967). In Experiment 1, we provide an analogous test case for geometric shapes by testing whether the same surface stimulus—a geometric shape made of line segments—can be assigned distinct parses in the geometry language of thought. If true, this would imply both that the mental representations of such shapes are structured and that efficiency is not the only factor underlying the choice among multiple structured representations (otherwise, the MDL representation would always be employed, regardless of other factors). We showed participants drawing animations depicting the generation of geometric shapes and tested whether the outcome geometric shape is encoded differently depending on the structure implied by the animation.

Methods

Transparency and openness

We report below how we determined our sample size, all data exclusions, all manipulations, and all measures in the study. The hypotheses, methods, and analysis plan were preregistered (https://osf.io/yrkzm/) prior to data collection.[1] The study materials (https://osf.io/thwec), anonymized data (https://osf.io/n6gte), and reproducible analysis scripts (https://osf.io/pkycr) are all publicly available on the OSF repository of the paper (Revencu et al., 2025). The data for all experiments were analyzed using R 4.4.0 (R Core Team, 2022), and the packages tidyverse 2.0.0 (Wickham et al., 2019), ggplot 3.5.1 (Wickham, 2016), rstatix 0.7.2 (Kassambara, 2023), lmerTest 3.1.3. (Kuznetsova et al., 2017), and emmeans 1.10.6 (Lenth, 2024).

Procedure

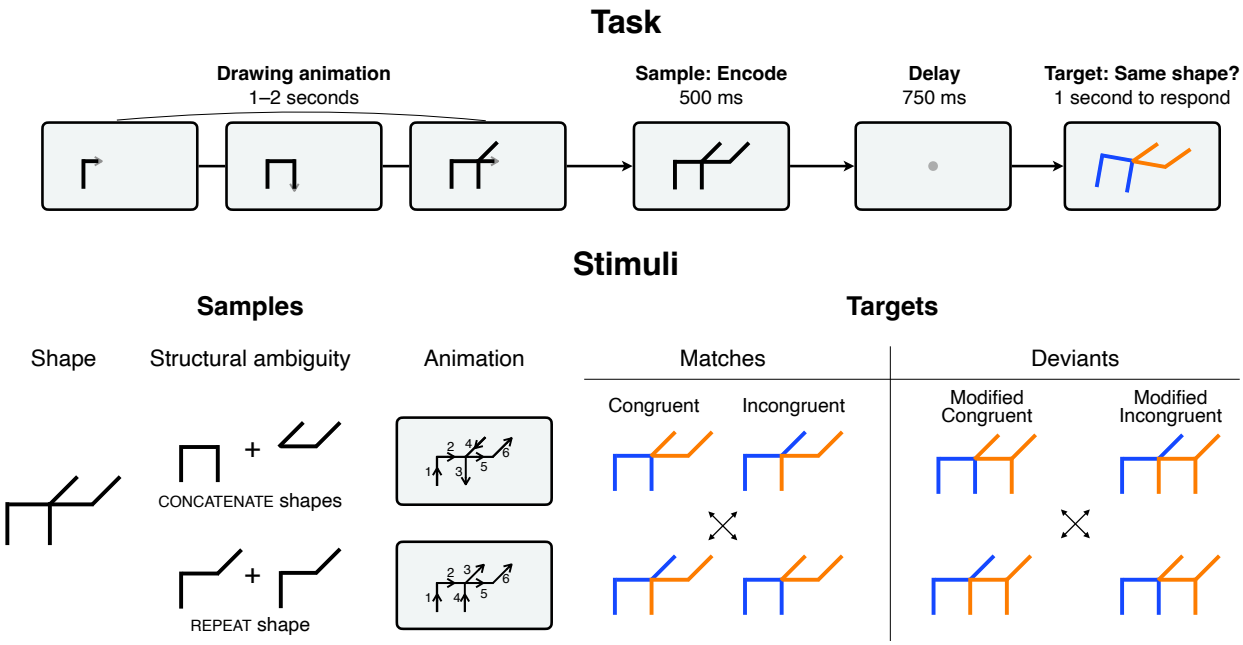
A schematic depiction of the task is shown in Figure 2 (top). Participants were engaged in a delayed same–different task: They had to decide whether two consecutive stimuli had the same shape, ignoring small changes in color, scale and orientation. Each trial started with an animation that generated a line drawing (1–2 seconds) by sequentially drawing two shapes (Segments 1–3 and Segments 4–6). A short

[1] As documented in Experiment 1's preregistration, we initially planned to recruit participants via social media. We set the sample size to 100, as this would have yielded a high-powered experiment even for a small effect size. Unfortunately, only 5 people completed the task in response to our post. We therefore decided to (i) recruit participants via Prolific instead; and (ii) run a pilot study to have a better estimate of the sample size needed to achieve high power. This pilot drove the sample size from 100 to 50.

break (333 ms) was interposed between the two shapes that formed the final drawing. The last frame of the animation depicted the **sample** image participants had to memorize and was shown for 500 ms before being removed. After a short break (250 ms), a fixation circle was displayed for 500 ms. The fixation circle was then replaced by one of the **target** images. Once the target image was displayed, participants used their keyboard to indicate whether the target had the same shape as the sample, ignoring differences in color, scale, and orientation. They were given 1 second to respond. If they made a mistake or did not respond fast enough, a feedback message was displayed for 1.5 seconds. In the experiment instructions, animations were presented in a neutral way: Participants were only told that they would have to remember a black-and-white shape generated by a drawing process. The experiment is available at https://neurospin-data.cea.fr/exp/Barbu/Shapes/experiment.html.

Figure 2

Experiment 1: Can two different structural representations be induced for the same shape?



Note. **Top**: Trial timeline, illustrated with a match congruent trial. Each trial started with a short animation that depicted the continuous generation of a shape. We illustrate the animation with a handful of frames and use gray arrows to depict drawing direction (not shown to participants). The final sample shape stayed onscreen for half a second. After a short delay, participants saw a new shape and had to decide whether it had the same shape as the sample, ignoring differences in color, scale, and orientation. **Bottom**: Illustration of the factorial design. Each sample shape could be drawn by two distinct animation programs. Similarly, each target shape could appear in two distinct coloring patterns. Trial congruity is defined by the alignment/misalignment between the structure implied by the sample animation and the target coloring. Deviants were created by adding a segment to the matches (shown on the right) or by moving a segment to a different part of the shape (not shown; see Figure 4).

Stimuli

Figure 2 (bottom) provides an overview of some of the visual stimuli used in Experiment 1. The sample shapes were 16 line drawings, generated with a version of the LoT language in Sablé-Meyer et al. (2022) implemented in Python. Each sample drawing could be generated by two different programs, either joining two identical shapes (REPEAT programs) or two different shapes (CONCATENATE programs). The programs were turned into animations that illustrated how a drawing was generated. For each sample drawing, the two animations—one corresponding to the REPEAT program, the other to the CONCATENATE program—were matched in duration and started from the same position. The length and width of the sample shape—the last frame of the drawing animation—varied between 90 and 180 pixels and was presented at the center of the screen in an invisible bounding box of 200 × 200 pixels.

For each sample image, we generated 24 colored target images: 8 matches and 16 deviants (Figure 2, bottom). The matches came in four color pairings (blue–orange, orange–blue, red–green, and green–red) and in two coloring patterns (REPEAT-congruent and CONCATENATE-congruent). The deviants were created in one of two ways: by *adding* a segment to the matches or by *moving* a segment of the matches to a different position. At test, the 24 targets could appear in one of four color pairings (blue–orange, orange–blue, red–green, and green–red), at different scales relative to the sample image (0.92, 0.94, 0.96, 0.98, 1.02, 1.04, 1.06, or 1.08) and in different orientations (−25°, −20°, −15°, −10°, 10°, 15°, 20°, or 25°).

Design and counterbalancing

The task was divided into two equal-length blocks. In each of the two blocks, every sample image was present in 16 trials, evenly divided into match and deviant trials. In the first block, half of the 16 sample images were assigned to the REPEAT animation program, half to the CONCATENATE animation program, randomly for each participant. In the second block, the assignments were reversed. Thus, all participants saw every sample shape animated by both programs an equal number of times over the entire experiment.

To generate the sample–target pairings in the first block, we randomly sampled a red-and-green color pair (red–green or green–red, specifying the color of the first shape) and a blue-and-orange one (blue–orange or orange–blue). This established the eight targets that would be shown in Block 1: For each sample image, there were 4 match targets shown in both color pairings, 2 add-segment deviant targets shown in one color pair, and 2 move-segment deviant targets shown in the remaining color pair. The eight targets were evenly divided into *congruent* and *incongruent* ones, depending on whether their coloring implied the same structure as the animation program of the sample, shown at the beginning of the trial. For each sample image, the 8 targets in a block were randomly assigned to a unique scale–orientation tuple.

Trials were presented in a pseudorandom order, such that each sample image appeared exactly once in every 16-trial sub-block. Block 2 followed the same logic as Block 1, except that it used the opposite animation programs for each sample image (i.e., a shape drawn with the REPEAT program in Block 1 would be drawn with the CONCATENATE program in Block 2), the opposite color pairings for the associated targets, and it re-randomized the scale–orientation assignments and the order of presentation. The randomization lists were generated online for each participant.

There were 256 test trials in total (16 sample images × 2 animation programs × 2 target coloring patterns × 2 target types × 2 color pairs). At the beginning of the experiment, participants completed a 16-trial Training block to get acquainted with the task. The Training block was identical to the main task but with different stimuli.

Participants

The final sample consisted of 50 English-speaking adult participants, recruited via Prolific. ($M_{\text{age}} = 31.5$ years, $\text{range}_{\text{age}} = 19$–$62$, collected via a free-response box). No other demographic information was collected. The sample size was chosen based on a pilot with 17 participants to detect an effect on response times with 80% power. An additional five participants were excluded due to not passing the inclusion criterion (detailed below). Participants gave informed consent before participating in the study and were paid 4 Euros for completing the task (median completion time: 23 minutes).

Data exclusion

As per the preregistration, we excluded the trials in which participants responded in less than 200 ms ($n = 20$ trials; 0.16%) or in which they provided no response ($n = 300$ trials; 2.34%). We also preregistered that we would exclude the participants who scored lower than 75% in accuracy or response rate ($n = 5$). Response-time data analyses are based on the trials where participants provided a correct response.

Results

Overall accuracy

Participants responded correctly on 87% of the trials, $t(49) = 58.85$, $p < .001$.

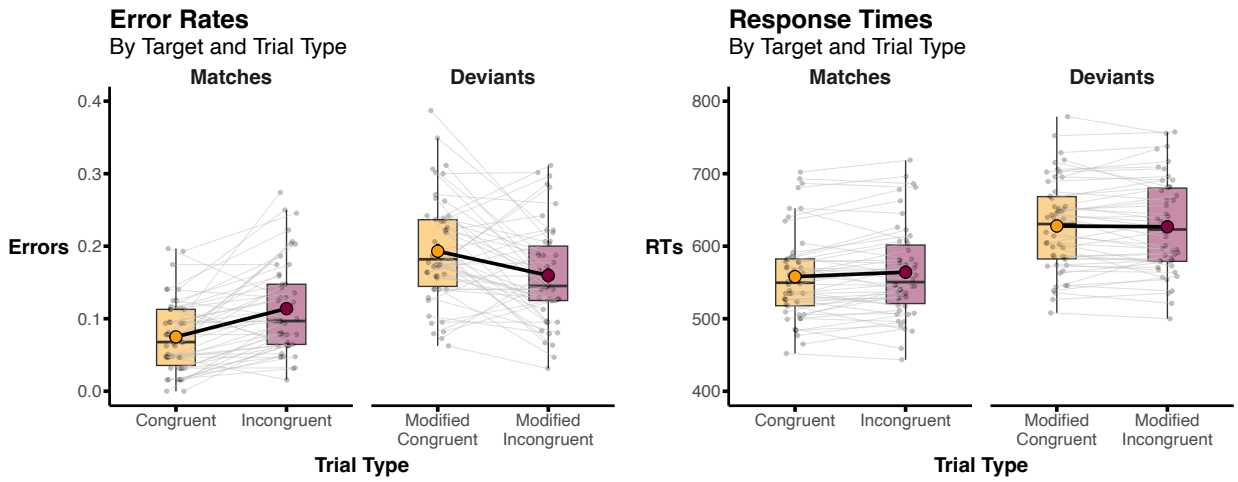
Main analysis: Match trials

As predicted, incongruent match trials, in which the target was colored according to a different structure from the animation, were harder than congruent match trials (Figure 3, bottom). Trial incongruity had a strong effect on error rates in the predicted direction, $M_{\text{difference}} = 0.04$, $t(49) = 5.42$, $p < .001$, 95% CI $[0.03, 0.06]$, Cohen's $d = 0.77$, and a small but nonsignificant effect on response times ($M_{\text{difference}} = 6.1$ ms,

$t(49) = 1.96$, $p = .056$, 95% CI $[-0.16, 12.34]$, Cohen's $d = 0.28$). The Bayesian t-test equivalents (Rouder et al., 2009) parallel the frequentist ones: decisive evidence for the alternative hypothesis in the case of error rates, $\mathrm{BF}_{10} = 9541.8$, and inconclusive evidence in the case of response times, $\mathrm{BF}_{10} = 0.89$.

Figure 3

Experiment 1: Shapes are easier to match when the animation and the coloring pattern point to the same structural interpretation



Note. Effect of congruity on error rates (left) and response times (right) by target and trial type. Gray circles and the lines connecting them represent individual averages as a function of trial type; colored circles depict group averages ±1 SEM; boxplots indicate the median and interquartile range.

Exploratory analysis (preregistered): Deviant trials

In the deviant condition, we found an effect in the opposite direction (Figure 3, bottom). Deviants which were formed by altering a congruent match gave rise to more false alarms than deviants which were modifications of an incongruent match, $M_{\text{difference}} = -0.03$, $t(49) = 3.13$, $p = .003$, 95% CI $[-0.06, -0.01]$, Cohen's $d = 0.44$. The Target Type × Trial Type interaction on error rates was also significant, $F(1, 49) = 28.97$, $p < .001$, $\eta_p^2 = .37$. As in the match condition, response times were not affected by trial congruity, $M_{\text{difference}} = -1.05$, $t(49) = 0.38$, $p = .71$, 95% CI $[-6.64, 4.54]$, Cohen's $d = 0.05$. Bayes factors point to the same conclusion: strong evidence for congruity effects on error rates, $\mathrm{BF}_{10} = 10.9$, and moderate evidence for the null hypothesis in response times, $\mathrm{BF}_{10} = 0.16$. Similar to the congruity effect on match trials, these findings indicate that a partial match in structural representations biased participants to respond "same", even when the correct response was "different". This strengthens the evidence for structural representation. Even for target shapes that were different from the samples, when their LoT representation were closer to the one induced by the animations, participants were more likely to misclassify them as matches.

Exploratory analysis (preregistered): Within-participant manipulation of shape representations

The overall results show that drawing animations induce corresponding representations of shapes *across* participants. Did we manage to induce distinct representations for the same shapes *within* participants as well? Recall that the two animation programs for each sample shape were blocked for each participant. Thus, in the first half of the experiment, any given shape was drawn by a single REPEAT or CONCATENATE program, which was switched in the second half of the experiment. One possibility is that the observed congruity effects stemmed primarily from the first half of the experiment, because the animations that participants saw first would dictate how they would represent these shapes throughout. This possibility would predict null or even opposite effects in Block 2. The other possibility is that participants could easily flip their interpretation of the same shapes according to the observed animation programs. If true, we should observe similar congruity effects in both experimental blocks.

To test this, we split the match trials by block and ran an ANOVA on error rates with block and trial type as factors. This analysis revealed a main effect of block (fewer errors in the second block: $F(1, 49) = 12.56$, $p = .001$, $\eta_p^2 = .2$), a main effect of trial type (more errors on incongruent trials: $F(1, 49) = 29.16$, $p < .001$, $\eta_p^2 = .37$), but no Block $\times$ Trial type interaction ($F(1, 49) = 2.75$, $p = .1$, $\eta_p^2 = .05$). Post-hoc t-tests indicated that the congruity effect was present in both blocks (Block 1: $M_{\text{difference}} = 0.05$, $t(49) = 4.15$, $p < .001$, 95% CI $[0.026, 0.074]$, Cohen's $d = 0.59$; Block 2: $M_{\text{difference}} = 0.03$, $t(49) = 3.78$, $p < .001$, 95% CI $[0.013, 0.042]$, Cohen's $d = 0.54$. Thus, the animation programs influenced the representation of the very same shapes within individuals across the two halves of the experiment.

Exploratory analysis: Can the results be explained by visual pop-out?

A potential alternative explanation to the results of Experiment 1 ascribes the congruity effect for matches to the fact that one of the target subshapes (as induced by the different colors) briefly appeared on-screen by itself in the middle of the animation phase, thereby providing a direct visual match.[2] We call this alternative interpretation "visual pop-out". Crucially, however, such a transient visual match was fully confounded with our congruity manipulation only in the match condition. In the deviant condition, target-match congruity and pop-out were partly dissociated, which allowed us to check whether the pop-out alternative can fully account for our results (Figure 4).

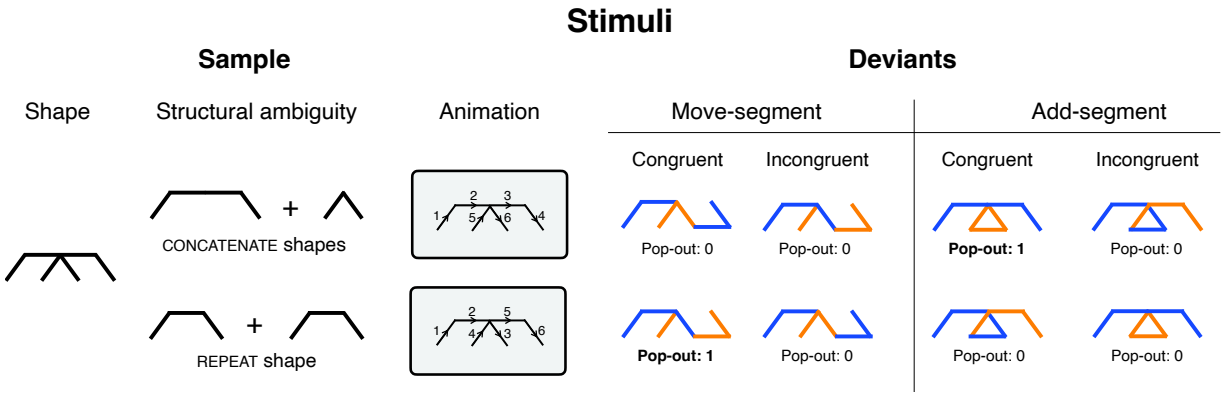
We exploited this partial dissociation to test whether pop-out is a better predictor than trial

[2] We thank an anonymous reviewer for raising this alternative interpretation.

congruity of error rates in the deviant condition. We coded each deviant–sample combination depending on whether the deviant coloring matched the shape that appeared by itself on screen in the middle of the animation period. For the sample shape depicted in Figure 4, for instance, the move-segment deviant congruent with the REPEAT program was coded as popout = 1, as the blue-colored subshape matched the shape that was presented on its own on the screen during the animation phase. By contrast, the MOVE-segment deviant congruent with the CONCATENATE program was coded as popout = 0, as the blue-colored subshape did not match the long shape briefly presented on its own during the animation phase.

Figure 4

Experiment 1: Testing an alternative "visual popout" interpretation of the results based on a transient match in the middle of the animation period



Note. Illustration of the popout coding for the deviants of one sample shape. If one of the colored subshapes matched the shape that was presented by itself during the animation phase, its popout value is 1; otherwise, 0.

We then ran three logistic regressions with error as the dependent variable, and congruity and/or pop-out as predictors, as well as random intercepts for individual subjects and stimuli. The Bayes Information Criterion (BIC) values ranked the congruity-only first (df = 4, BIC = 5202.41) and the combined model second (df = 5, BIC = 5210.61), while the popout-only model came in last (df = 4, BIC = 5213.63). Moreover, in the combined model, which controls for the popout interpretation, trial congruity continued to significantly predict error rates, $\beta_{\text{congruity}} = -0.31, \text{SE} = 0.09, p < .001, 95\%\text{CI} = [-0.48, -0.13]$, whereas the pop-out variable did not $\beta_{\text{pop-out}} = 0.08, \text{SE} = 0.1, p = .455, 95\%\text{CI} = [0.28, 0.12]$. These results rule out the visual popout alternative for the congruency effect, at least for the deviant trials. But if the effect is driven by the same mechanism on both match and deviant trials, this alternative is ruled out for the experiment as a whole. That said, future work could refute visual popout more directly by deconfounding transient visual matches

and shape structure on match trials as well.

Discussion

By using a geometric equivalent of ambiguous sentences, Experiment 1 provides evidence that adults encode geometric shapes as structured representations. Just as the same sequence of words can have two distinct syntactic parses, the same geometric shape can have two distinct mental programs. This implies that the preference for minimum description length representations of geometric shapes is a soft constraint that can be easily overcome. Otherwise, participants would have eventually settled on the most efficient configuration of each shape—in our stimulus set, this corresponds to the REPEAT configuration, which achieves a higher degree of compression than the CONCATENATE one. While it may seem intuitive that observing the drawing process will influence the representation of the outcome shape, this does not follow, without further qualifications, from minimum description length accounts, which ascribe a primary role to simplicity and efficiency in mental representation. By contrast, the results of Experiment 1 indicate that we successfully manipulated the representations used to encode the very same shape both across and within participants. The perceptual system may be constantly searching for the most compressed representation of what it perceives, but this minimum may not be so easy to identify. Indeed, the reader can verify with the shapes in Figure 2 and Figure 4 that, in the absence of a biasing animation, these shapes may not be so easily interpreted as a repetition of two identical subshapes.

Previous work has shown that different parses of identical stimuli can emerge in a top-down manner either via different visual contexts or via category learning, which often creates novel visual primitives (Goldstone, 2000; Schyns et al., 1998). The present Experiment 1 complements this body of work in at least two ways. First, we introduce a new Gestalt-like cue to grouping—the drawing animation—that we derive directly from the semantics of the shape LoT, whose primitives correspond to implicit motor commands (e.g., TRACE a segment, TURN the drawing angle, etc.). Second, recall that our manipulation worked not only across participants but also within participants. The second block, in which all drawing animations flipped from one version to the other, gave rise to the same congruity effects as the first block. This suggests that the drawing animation has a transient influence on the representation of the outcome shape, rather than the longer-term effects produced by the formation of new primitives during the learning of a novel task.

Experiment 2: Subtree Facilitation

In Experiment 2, we explored the mental representation of geometric shapes with a different constituency test. Inspired by the finding that linguistic strings are remembered better if they belong to the same subtree (Jarvella, 1970, 1971), we ask whether adults are more likely to recognize a subpart of a shape

when the subpart corresponds to a subtree in the shape's LoT representation. This paradigm is closely related to the hidden-figure test in the Gestalt psychology tradition (Leeuwenberg et al., 1985; Reed & Johnsen, 1975) and in particular to Experiment 3 in the seminal work of Palmer (1977), but departs from it in several key respects. First, our focus on the shape language of thought moves us beyond the Gestalt theory of grouping. The shape LoT is a much more general model that generalizes to a much broader range of stimuli than the simple shapes used in previous work (e.g. Palmer, 1977). Moreover, we designed the stimuli to minimize Gestalt cues to parsing (e.g., no stimulus could be parsed into subtrees based on closure, which is a strong cue to objecthood), thus distinguishing our predictions from those derived from Gestalt principles. Second, the sub-shapes we investigated did not correspond to subtrees at the first level of the hypothesized representation but at the second one—they were subtrees of subtrees. Finding subtree effects at a deeper level in the tree is better evidence of hierarchy, since evidence confined to the first level would be compatible with nonhierarchical segmentation as well. Third, we use a rigorously controlled set of stimuli, fully orthogonalized within participants, which ensured that any observed effect would be due solely to the relation between the sample shapes and their subparts.

Methods

Transparency and openness

The hypotheses, methods, and analysis plan were preregistered prior to data collection (https://osf.io/k2cx8/). Research materials (https://osf.io/6wgam), anonymized data (https://osf.io/4pfsk), and analysis scripts (https://osf.io/98gbv) are publicly available.

Procedure

An overview of the task can be found in Figure 5 (top). Participants were engaged in a delayed match-to-sample task. On each trial, participants saw a static sample shape and, after a delay, decided whether a smaller target shape was a subpart of it. Each trial started with a fixation circle presented at the center of the screen for 500 ms. The fixation circle was then replaced by a sample line drawing. The drawing stayed on the screen for 3 seconds for participants to memorize it. After a short break (200 ms), the fixation circle was displayed again for another 500 ms. Then, a smaller target image consisting of 2 conjoined segments was flashed 200 pixels up or down of the fixation circle for 300 ms. (This short presentation time was meant to minimize eye movements.) Both the fixation circle and the smaller image then disappeared, and subjects had to indicate on their keyboard, within one second, whether the target was a subpart of the sample. If they responded incorrectly or failed to respond, participants were presented with a feedback message for 2 seconds. Then, the next trial started. The experiment is available at

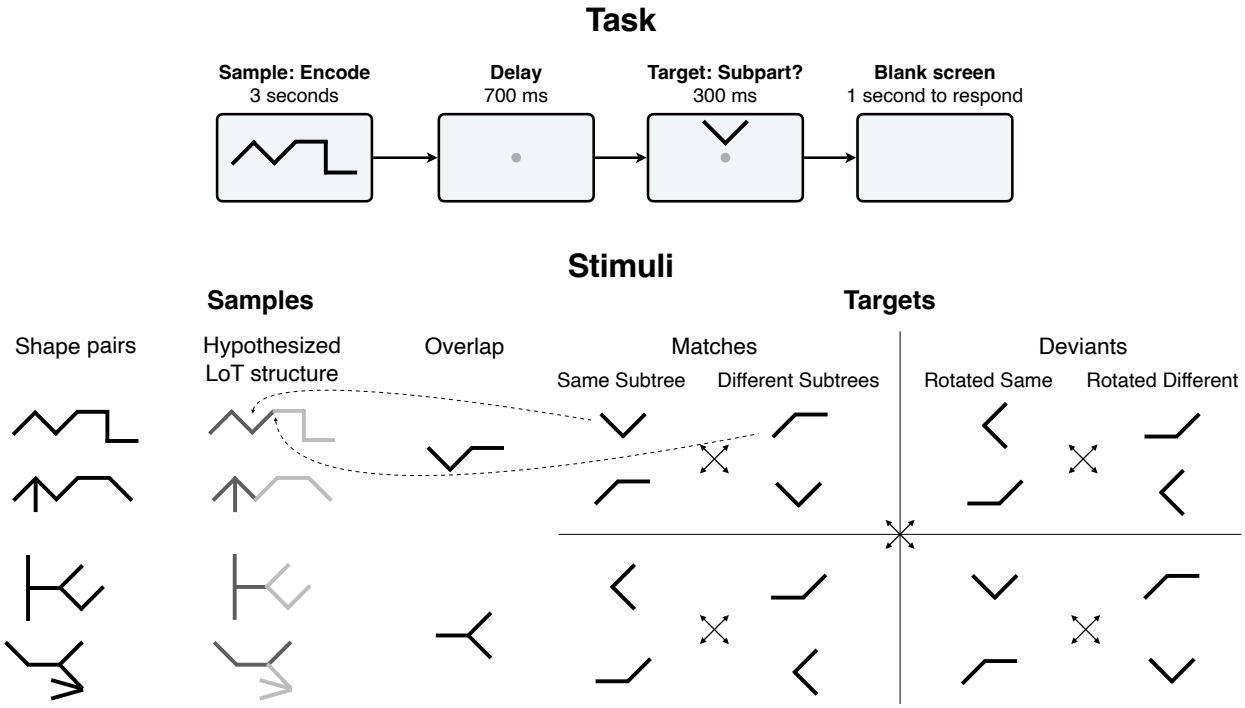
https://neurospin-data.cea.fr/exp/Barbu/Shapes4/experiment.html.

Stimuli

As in Experiment 1, we created 16 sample images using the language in (Sablé-Meyer et al., 2022). Each image consisted of 6 equal-length segments and was generated by concatenating two subshapes of 3 equal-length segments each. In turn, each subshape was created by the two signature operations of the LoT language by Sablé-Meyer et al. (2022): repetitions (e.g., of a program n times) and subprograms (e.g., execute a program, then return to the original position).

Figure 5

Experiment 2: Are subparts of a large shape easier to find when they coincide with a subtree in the internal structured representation?



Note. **Top**: Trial timeline, illustrated with a Match/Same-subtree trial. Participants decided whether the second shape (target) was part of the first one (sample). **Bottom**: Example of a stimulus group in Experiment 2. Each sample shape (left) was created by concatenating two subshapes, each of which was built using the REPEAT operation in the shape LoT. Because this operator leads to high compression, we hypothesized that participants would parse the large shapes into the two subshapes, as indicated by the coloring to the right of the samples (not shown to participants). Each pair of shapes had three segments in common, which were used to create two 2-segment matches. For each shape, one of the two matches was a same-subtree match: Its segments belonged to a single subshape in the hypothesized LoT structure; the other subshape was a different-subtree match, because its segments spanned distinct subtrees in the LoT structure. The same-subtree target for one stimulus was a different-subtree target for the other, and vice versa. The deviant targets were formed by rotating the match targets. The matches for one stimulus pair (the top two sample shapes on the left) were the deviant targets for another stimulus pair (the bottom two sample shapes on the left).

The 16 images were generated to form four groups of four images each (see Figure 5, bottom, for an example of a 4-image group). Each of the four groups consisted of two pairs of images. In each group, the images in the first pair were designed to contain a common 3-segment connected subshape. This subshape was used to generate two target match images (left segment + center segment and center segment + right segment). For every pair of images, the two match images were present in both, but they differed regarding their status in the underlying program structure: One image aligned to the tree-structure of the program (same subtree), whereas the other did not (different subtrees). Some of the segments that formed the subshape were connected to other segments, and some were not; this was also matched for the common subshapes in each pair.

We then created two deviant targets by rotating the two match targets by 90, 180, or 270 degrees. These targets were concatenated to create another 3-segment connected subshape, which was used as the basis for the images in the second pair of each group. This allowed us to perfectly balance the targets, as the match targets for one stimulus pair were the deviant targets for another. This orthogonal design ensured that no property other than the *structural relation* between sample and target could account for the expected experimental effects.

Note that the generative process used for the 6-segment shapes does not formally guarantee that the programs we used correspond to the shapes' true MDL programs. The exponential program space and high complexity of the stimuli made it intractable to identify the shortest program with certainty. We addressed this limitation in three ways. First, we ensured that the two concatenated subshapes were optimally compressed by having their programs embedded under the REPEAT operator, which affords the highest degree of compression in our shape language. Second, we made sure that different splits of the 6-segment shapes (e.g., into three groups of two segments, or into one group of four and another of two) could not use the REPEAT primitive to the same degree. Third, our hypothesis did not require perfect structural alignment—it was sufficient that participants' mental representations aligned with our hypothesized structure only probabilistically.

Design and counterbalancing

Unlike Experiment 1, the targets did not vary in color, scale, or orientation. However, to prevent the possibility of a direct physical match between the sample and target, the targets were flashed above or below the central position where the sample image had been presented. Within each pair, identical targets were flashed at identical positions. Also, within each pair, each target appeared exactly twice in each of the two possible positions (above or below fixation) and for each of the two orientations. To have the average distance between the 3-segment common subshapes and fixation equal 0, we also presented each

sample-target combination rotated by 180 degrees. Within each pair, each target appeared exactly twice in each of the two possible positions (above or below fixation) and for each of the two orientations. Finally, the 3-segment subshape that was common to a stimulus pair was shown at the same position on participants' display. To find the display coordinates for each pair, we centered both images in the same bounding box, then moved them towards each other until the common subshape targets were overlapping.

There were 128 test trials in total (16 sample images × 4 targets × 2 orientations). The order of trials was pseudorandomized such that any given sample image could not appear more than once every 16 trials. As in Experiment 1, before the main task, participants completed a 16-trial Training block with different stimuli, which emphasized that *rotated* subpart targets (i.e., deviants) must be rejected as subparts. A demo version of the experiment is available at https://neurospin-data.cea.fr/exp/Barbu/Shapes4/experiment.html.

Participants

The final sample consisted of 50 English-speaking adult participants, recruited via Prolific ($M_{\text{age}} = 28$ years, $\text{range}_{\text{age}} = 18$–66, collected via a free-response box). No other demographic information was collected.). The sample size was chosen based on a pilot with 9 participants to detect an effect on response times with 80% power. Four additional adults were tested but not included in the final sample because they failed to meet the inclusion criterion (detailed below). Participants gave informed consent before the task and were paid 3.5 Euros for completing it (median completion time: 20 minutes).

Data exclusion

As preregistered, we excluded from the final analysis the trials in which no response was provided ($n = 482$ trials, 7.5%), as well as the participants who provided responses on less than 50% of the match trials ($n = 4$, 3 of whom did not provide an answer to any of the 128 trials). We did not preregister any accuracy threshold, because we knew from the pilot that the task would be difficult—we return to the relation between task difficulty and our measure of interest in the Results section. Response times were measured from target onset, but fast-response trials were no longer an issue, as participants were anyway not allowed to respond in the 300-ms time window during which the target was displayed. Response-time analyses only take correct trials in consideration.

Results

Overall accuracy

Participants solved the task significantly above chance level (mean accuracy 55%, $t(49) = 4.42$, $p < .001$), although the relatively low accuracy rate indicates that this task was much harder than in Experiment 1.

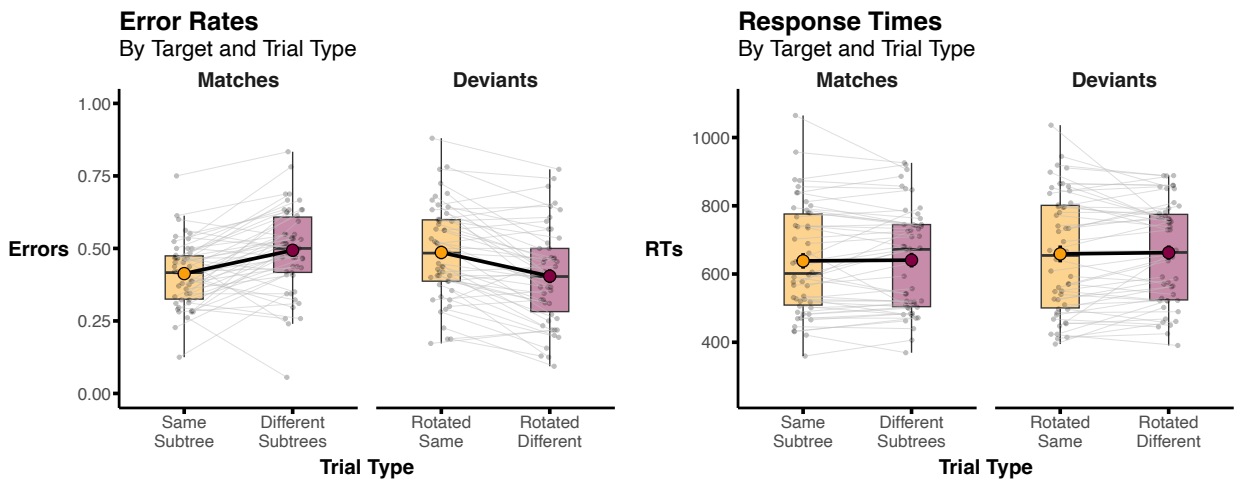
We return to this issue after presenting the main experimental effects.

Main analysis: Match trials

As predicted, same-subtree match trials were easier than different-subtree match trials (Figure 6). The subtree-facilitation effect was not present in response times ($M_{\text{difference}} = 2.23$ ms, $t(49) = 0.27$, $p = .787$, 95% CI $[-14.2, 18.7]$, Cohen's $d = 0.04$)[3] but was robust on error rates ($M_{\text{difference}} = 0.08$, $t(49) = 4.59$, $p < .001$, 95% CI $[0.05, 0.12]$, Cohen's $d = 0.65$). Bayesian t-tests revealed decisive evidence in favor of the alternative hypothesis for error rates, $\text{BF}_{10} = 672.73$, and moderate evidence for the null for response times, $\text{BF}_{10} = 0.16$.

Figure 6

Experiment 2: Two line segments are easier to find within a shape when they belong to the same subtree



Note. Effect of congruity on error rates (left) and response times (right) by target and trial type. Gray circles and the lines connecting them represent individual averages as a function of trial type; colored circles depict group averages ±1 SEM; boxplots indicate the median and interquartile range. Response times measurements started at target onset.

Exploratory analysis: Deviant trials

In the deviant condition, we found an effect in the opposite direction, just like in Experiment 1. Deviants which were rotated same-subtree matches gave rise to more false alarms than the deviants which were rotated different-subtree matches, $M_{\text{difference}} = -0.08$, $t(49) = 5.71$, $p < .001$, 95% CI $[-0.11, -0.05]$,

[3] Given the relatively low accuracy in this experiment, an analysis limited to correct trials could be unrepresentative. Therefore, at a reviewer's suggestion, we re-analyzed response times using all trials (correct and incorrect). We found no substantive change in the results: $M_{\text{difference}} = 3.38$ ms, $t(49) = 0.57$, $p = .571$, 95% CI $[-8.53, 15.29]$, Cohen's $d = 0.08$.

Cohen's $d = 0.44$. The Target Type × Trial Type interaction on error rates was also significant, $F(1, 49) = 42.63$, $p < .001$, $\eta_p^2 = .46$. As in the match condition, we observed no effect on response times, $M_{\text{difference}} = 3.93$, $t(49) = 0.38$, $p = .704$, 95% CI $[-16.75, 24.62]$, Cohen's $d = 0.05$. Bayes factors again indicated decisive evidence for a congruity effect on error rates, $\text{BF}_{10} = 24378.3$, and moderate evidence for the null hypothesis for response times, $\text{BF}_{10} = 0.16$. We had no expectation about participants' responses on deviant trials, but the inverse subtree facilitation effect on deviant trials is consistent with the main hypothesis of the study. It indicates that participants recognized the subtrees even under the rotated deviant guise, which made them more likely to incorrectly report them as subparts of the sample images. This suggests that shape LoT representations abstract away from superficial details such as orientation.

Exploratory analysis: Does participants' low accuracy jeopardize the interpretation of the results?

The low overall accuracy of participants in Experiment 2 may raise doubts about the validity of the results. In addressing this legitimate worry, let us start by pointing out that the low performance was a consequence of the high complexity of the stimuli: On average, the description length of the hypothesized programs was 19, a much higher value than in Experiments 1 and 3 and than in previous work (Sablé-Meyer et al., 2022). In turn, this high complexity was required to obtain a perfectly balanced stimulus set, in which the congruent matches for one sample were incongruent matches for another (and vice versa) and in which the matches of one sample were the deviants of another (and vice versa).

Second, performance level is, in fact, orthogonal to the effect of interest. We predicted that, regardless of how difficult the task would be, same-subtree matches should be easier to detect as subparts than different-subtree matches. Not only did this prediction turn out correct, but its corollary was confirmed for deviant targets as well: Rotated same-subtree deviants were more unlikely to be correctly rejected than rotated different-subtree ones.

Third, we found that this result held even when we split our sample of participants into those that performed above chance and those who did not. To obtain this partition, we ran a one-sided binomial test for each participant (k = number of successes, n = number of valid trials, H_0: $p = 0.5$; H_1: $p > 0.5$) and split participants into two groups, depending on the p-value of the binomial test: high performance (p-value $< .05$) and low performance (p-value $\geq .05$). The effect on error rates was present in both groups and for both matches and deviants (Table 1).

Another way to think about the relation between accuracy and our predictions is via signal detection theory. While participants' overall sensitivity was indeed low (even if significantly different from 0: $d' = 0.27$, $t(49) = 4.45$, $p < .001$), their bias varied depending on the subtree status of the target. For

Table 1

Error rate effects (incongruent − congruent) by performance level in Experiment 2

PERFORMANCE	N	TARGET TYPE	ERROR RATE DIFFERENCE	t-VALUE	p-VALUE	95% CI	COHEN'S d
High	12	Match	0.13	3.68	**.004****	$[0.05, 0.21]$	0.53
		Deviant	−0.09	−3.23	**.008****	$[-0.15, -0.03]$	0.76
Low	38	Match	0.07	3.28	**.002****	$[0.02, 0.11]$	1.06
		Deviant	−0.08	−4.71	**<.001*****	$[-0.11, -0.05]$	0.93

Note. Output of t-tests on error rates target type and performance levell. Significant p-values are written in boldface.

congruent trials, participants' bias was negative ($c = -0.1$, $t(49) = -2.4$, $p = .02$), indicating a more relaxed criterion for answering "yes, this target is a subpart", whereas for incongruent trials, the bias was positive ($c = 0.12$, $t(49) = 2.24$, $p = .029$), indicating a stricter criterion for answering "yes". Thus, while the participants struggled with the distinction between matches and deviants, they were highly sensitive to the distinction between congruent and incongruent samples in their decision-making. And it was this latter distinction, not the former, that our manipulation targeted. Nevertheless, we acknowledge that future replications of this experiment would benefit from increasing the encoding window, in order to boost participants' sensitivity to the distinction between matches and deviants.

In sum, the low overall accuracy was orthogonal to the directional effect of interest, both by design and in the data. The congruity effect was present for both match and deviant targets, regardless of individual performance, and closely paralleled the pattern of results in Experiment 1.

Discussion

Experiment 2 provides further evidence for constituent structure in geometric shape representations, as participants were more likely to detect same-subtree segments as matches than different-subtree ones. Experiment 2 also shows that deviants created by rotating same-subtree matches are more likely to be (incorrectly) reported as matches than different-subtree rotations. This novel finding suggests that the tree representations assigned to the shapes are highly abstract, as they do not encode the highly salient feature of orientation. Nevertheless, it remains possible that geometric shapes are not represented as nested tree structures but as concatenated chunks (e.g., Orbán et al., 2008). Even though we tested *subparts* of subtrees, a sequential process of initial chunk segmentation followed by a within-chunk template matching might also account for our results. In Experiment 3, we test this alternative by probing the hierarchical tree structure of the underlying representations in a more fine-grained manner.

Experiment 3: Syntactic Depth

In Experiment 3, we draw inspiration from *wh*-movement in linguistics. A characteristic of natural language is that phrasal constituents can be moved, for instance in the generation of relative sentences (*I saw the funny dog* → *The funny dog which I saw* _), whereas non-constituent sequences of words cannot (**The funny which I saw* _ *dog* is invalid) (Haegeman, 2006). Furthermore, psycholinguistic evidence indicates that object extraction takes longer than subject extraction in languages with a strict Subject–Verb–Object order, such as English (for a review, see Gibson, 2001). For instance, *The man who John met* _ *was nice* is more difficult to process than *The man who* _ *met John was nice*, as the moved constituent *The man* in the first sentence has to travel a longer distance from its original (object) position to the sentence-initial one. Here, we test whether the ease with which shapes are reconfigured depends on the depth of the tree at which the shape is split. Once again, on each trial, participants were exposed to a sample line drawing that they had to memorize. They were then presented with an image that split the original shape into two parts and they had to decide whether the segments in the new images could be rearranged to form the original image. Unbeknownst to the participants, the point where the sample shape was split varied in terms of the tree level at which the cut was made. If participants mentally traverse a tree structure when recombining the shapes to check if they match, we should find an effect of tree structure and tree depth on participants' behavior. We predicted that reassembling two fragment shapes would be easier when those fragments split the original shape's tree structure at its joints than when they do not, and all the easier when the split occurs higher up in the tree structure.

Methods

Transparency and openness

The hypotheses, methods, and analysis plan were preregistered prior to data collection (https://osf.io/czub7/).[4] Research materials (https://osf.io/r74dq), anonymized data (https://osf.io/us3fp/), and analysis scripts are publicly available (https://osf.io/skr7u).

[4] We initially planned a different version of Experiment 3, which would have tested whether reorienting attention across distinct segments of a shape depends on the shape's tree structure. In preparation of the planned Experiment 3, we ran a preliminary object-attention paradigm using the stimuli in Experiment 2 (https://osf.io/yvapg/). The null results we found indicated that LoT representations may not be automatic and prompted us to devise another paradigm for Experiment 3. As automaticity was orthogonal to the current study, we decided to look into the automaticity of LoT representations more systematically in a different study.

Procedure

An overview of the task is shown Figure 7 (top). Participants had to remember a sample image, then decide whether the two fragments of the target image could be rearranged by a simple translation to recreate the memorized image. Each trial started with a central fixation dot displayed for 500 ms. The fixation dot was replaced by the sample line drawing, which was displayed for 2 seconds. A mask was then presented for 200 ms to prevent participants from relying on iconic memory. After a 500-ms blank screen, the fixation dot was again displayed for 500 ms. The target image was flashed for 200 ms, at the same central position as the sample. Afterward, participants had 1.5 seconds to decide whether the two fragments in the target image could be reconfigured to form the sample image. Compared to Experiments 1 and 2, we extended participants' response time by 500 ms, hypothesizing that the null effects on response times in Experiments 1 and 2 were due to a speed–accuracy tradeoff induced by time pressure: Participants may have rushed their responses to avoid missing the response window. If they responded incorrectly or failed to respond, participants were presented with a feedback message for 1 second. After a 500-ms break, the next trial began. The experiment is available at https://neurospin-data.cea.fr/exp/Barbu/MovementMask/experiment.html.

Stimuli

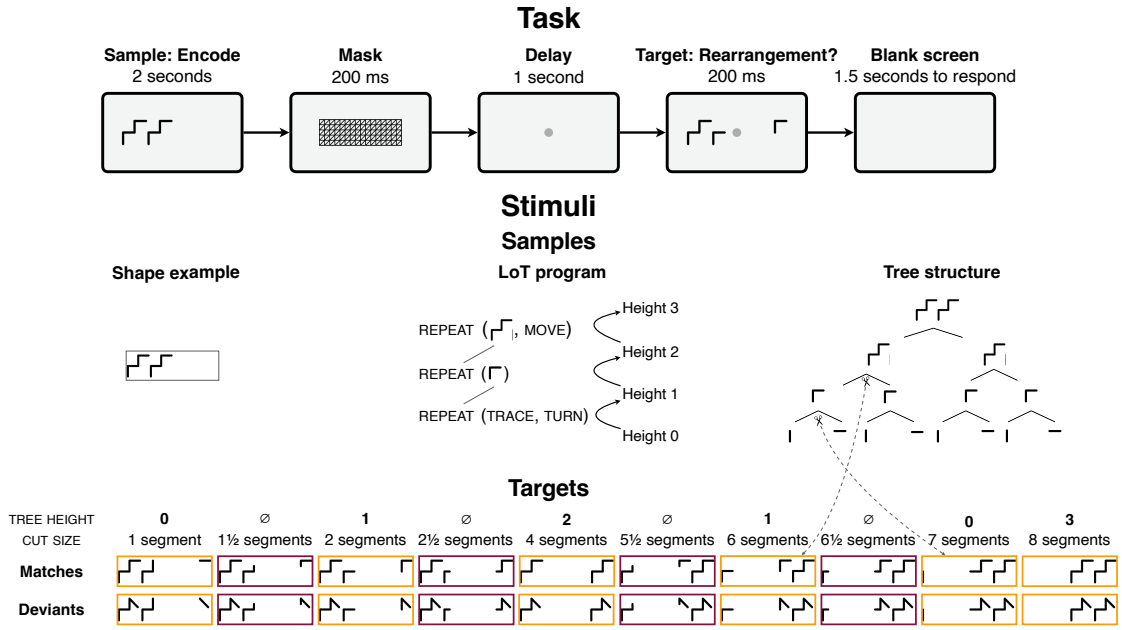
We created 8 sample stimuli using the same geometry language as in Experiments 1 and 2. Each sample consisted of three nested binary repetitions and thus comprised 8 segments that could be parsed into three different hierarchical levels. The sample illustrated in Figure 7 (middle), for instance, was generated by the following program: REPEAT{ REPEAT[REPEAT(TRACE, TURN)], MOVE} . The innermost REPEAT operator creates the 2-segment |‾ subpart; the central REPEAT operator duplicates it to create a 4-segment subshape; and the outermost operator duplicates the 4-segment subshape to create the final shape. As shown in Figure 7 (middle), each application of the REPEAT procedure produces one more level of hierarchy in the shape's tree.

The samples fit a bounding box of 190 (width) × 120 (height) pixels, but were left-aligned in a bounding box twice the width. This allowed us to display both the samples and the targets in an invisible bounding box of 380 × 120 pixels. Each left-aligned sample–target combination was also rotated by 180 degrees to create a symmetric right-aligned sample–target combination. This balanced the distance between the central fixation point and the sample positions across trials.

To create the match targets, we took the sample as our starting point and moved chunks of different sizes (1, 1.5, 2, 2.5, 4, 5.5, 6, 6.5, 7, or 8) to the other end of the 380 × 120 invisible bounding box (Figure 7, bottom). This allowed us to ask whether the difficulty of checking that two fragments can be

Figure 7

Experiment 3: Does the tree structure of a shape predict how easy it is to reassemble from two of its fragments?



Note. **Top**: Trial timeline, illustrated with a Match/Tree structure trial. Participants decided whether the two fragments shown as target could be reassembled to form the original sample shape. **Middle**: Example of a sample shape, together with its LoT generative program and tree structure. **Bottom**: The targets consisted of 10 matches, which preserved the sample's tree structure (orange outlines), and 10 deviants, which did not (purple outlines). For each target, we indicate the tree height at which the sample was cut (a numerical value for tree-preserved targets, an empty value for tree-broken targets, as well as the size of the cut chunk (decreasing from left to right). Dashed arrows illustrate the link between cuts in the sample's tree and the resulting targets. Outline colors and bounding boxes are for illustrative purposes and were not shown to participants.

put together to form an original shape depends on where the cut is made. To create the deviant targets, we first modified the samples by changing the orientation of one or two segments in the original drawing. We then modified this deviant sample to create targets exactly as for the match sample.

Design and counterbalancing

The match targets were divided into two conditions. In the tree-preserved condition, the independent variable was the tree level at which the cut was made, which partly overlapped with the chunk of the moved size. If the entire sample image moved (e.g., from left to right), this would correspond to a cut at the highest level of the tree (height 3), as well as to a movement of size 8. Analogously, moving four segments corresponded to a cut at height 2; moving two segments corresponded to a cut at height 1; and moving one segment corresponded to a cut at height 0 (the lowest level of the tree).

Up to this point, tree height and the size of the moved chunk are perfectly aligned: Taking the

logarithm to the base 2 of the size of the cut outputs the tree height of the cut. We addressed this issue in two ways. We first dissociated tree height from the size of the moved chunk by introducing movements of size 6 (corresponding to height 1 and thus equivalent to movements of size 2 from a tree-based perspective) and movements of size 7 (corresponding to height 0 and equivalent to movements of size 1). We then further deconfounded these two variables by introducing a tree-broken condition, in which the size of the moved chunk also varied parametrically (1.5, 2.5, 5.5, and 6.5) but which always disrupted the tree structure. The deviant targets followed the same logic. Note that the tree-depth account makes the counterintuitive prediction that the higher the pixel overlap between the sample and target, the more difficult it should be for subjects to answer the test question (Figure 7).

Across all stimuli, the possible sizes of the moved chunks were 1, 1.5, 2, 2.5, 4, 5.5, 6, 6.5, 7, and 8. Because moving 2 segments is equivalent in terms of tree structure to moving 6 segments (and the same for 1 and 7), half the stimuli received 6 and 1 as targets for one orientation, and 2 and 7 for the other (and vice versa). The order of presentation was pseudo-randomized. Samples were presented in cycles such that no shape could appear more than once in a block of eight trials. Within each sub-block, shape order was randomized. Each sample–target pairing was presented only once.

The procedure consisted of 256 trials in total [8 shapes × 2 orientations × 4 conditions (match, match-control, deviant, deviant-control) × 4 movement levels]. At the beginning of the experiment, participants completed a 16-trial Training block to get acquainted with the task. The Training block was identical to the main task but used unrelated stimuli (e.g., a triangle).

Participants

The final sample consisted of 36 English-speaking adult participants, recruited via Prolific ($M_{\text{age}} = 30$ years, $\text{range}_{\text{age}} = 18$–56), collected via a free-response box). No other demographic information was collected. The sample size was chosen based on a pilot with 11 participants. The pilot data indicated that 50 subjects would suffice to detect the predicted main effects and interactions on response times with >95% power. Fourteen additional adults were tested but not included in the final sample due to their failure to meet the inclusion criterion ($n = 13$) or technical error ($n = 1$: their data were not stored on the server). As pilot data indicated that even with 36 subjects, statistical power exceeded 90%, we did not replace excluded participants. All participants gave informed consent before participating in the study and were paid 5 Euros for completing the task (median completion time: 27 minutes).

Data exclusion

As per the preregistration, we excluded the trials in which participants failed to provide a response ($n = 94$ trials; 1.02%) and the participants who scored below 56% accuracy, which corresponds to the above-chance

level for 256 trials at $\alpha = .05$, or below 70% response rate ($n = 13$). As in Experiment 2, response times were measured starting from target onset. We therefore did not need to exclude fast-response trials, since participants were not allowed to respond in the 200-ms time window during which the target was flashed. Response-time analyses were based on the trials where participants answered correctly. Error rate analyses were based on the trials in which participants provided a response.

Results

Overall accuracy

Participants responded correctly on 75% of the trials, $t(35) = 13.78$, $p < .001$.

Main analysis

We preregistered three main hypotheses. First, we predicted that the higher up in the tree the chunk was cut out of the sample image, the easier it would be for participants to detect that a match is a rearrangement of the original image. Second, we predicted that this effect would be due to the tree-structured representation of the target, not merely to the size of the moved chunk. We therefore predicted a Chunk-Size × Condition interaction in the match trials. Third, we predicted an interaction in the tree-preserved condition between Chunk-Size effect and Target type (match versus deviant), as we expected a stronger effect in the match compared to the deviant condition.

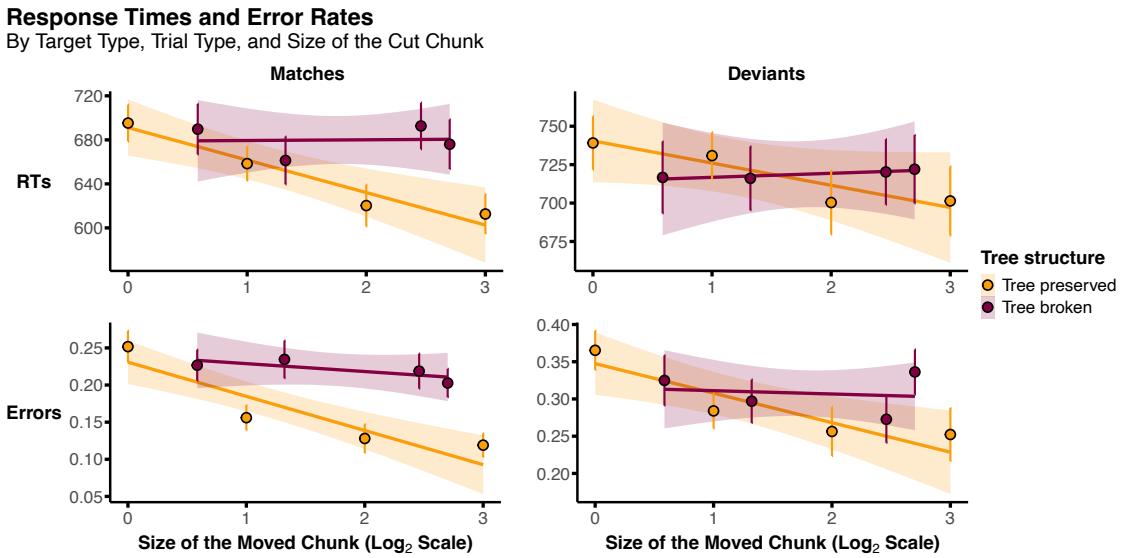
To test these predictions, we averaged the data by subject, tree structure, and the logarithm (base 2) of the size of the moved chunk. In the tree-preserved trials, the $\log_2$-size is equivalent to the tree height at which the shape was cut. Because our main hypothesis concerned participants' responses to the tree-preserved matches, this analysis treats size-6 movements equivalently to size-2 movements (height-1 splits), and size-7 movements equivalently to size-1 movements (height-0 splits). We return to this in the next subsection. In the tree-broken trials, the $\log_2$-size does not have any tree equivalent, since the tree structure is broken in all cases where a segment is split halfway. This gave us 16 response times and 16 error rates for each participant [4 chunk-sizes × 2 conditions (tree preserved versus tree broken) × 2 target types (match versus deviant)].

We standardized both dependent variables to boost the interpretability of the regression coefficients and created a new variable, *condition*, for each of the four levels in our design (Match versus Deviant × Tree Preserved versus Tree broken). We then fit the preregistered linear mixed model $z_{\mathrm{RT}}|z_{\mathrm{Errors}} \sim \log_2 \text{chunk-size} * \text{condition} + (1|ID)$ to the data. As the tree-preserved match condition was set as the reference level, we could test all three predictions with a single model: the effect of tree height in the tree-preserved match condition and the two interactions. In addition, the coefficients of the model were

used to obtain the marginal means estimates for the influence of $\log_2$-chunk size for each condition separately, which we report below. The model also accounted for individual differences by including a random intercept for each subject. Participants' behavioral results are plotted in Figure 8, while the results of the linear mixed model are shown in Table 2.

Figure 8

Experiment 3: Two fragments of a shape are easier to reconfigure when they are split higher up in the tree



Note. Effect of the size of the moved chunk ($\log_2$ scale) in match and deviant trials on response times (top) and error rates (bottom). Colored circles represent group averages, error bars represent ± 1 SEM, and shaded intervals represent 95% confidence intervals of the regression line. Values on the x-axis are log-scaled. To align integer values on this scale with the different levels in the tree structure (0 = deepest level; 3 = highest level), we grouped cuts of sizes 1 and 7 (0 on the x-axis) and cuts of sizes 2 and 6 (1 on the x-axis).

The response-time results confirmed our predictions (Figure 8, top). Participants found it easier to reconfigure the two parts shown as targets the higher up in the tree they were split ($\beta_{\text{Tree-Preserved Matches}} = -0.22$, $SE = 0.03$, $p < .001$, 95% CI $= [-0.28, -0.15]$). The two control conditions showed that this effect was due to tree structure and not to other low-level features. There was an interaction between $\log 2-$Chunk size and Tree structure in match trials ($\beta_{\text{Match Preserved} \times \text{Broken Interaction}} = 0.22$, $SE = 0.06$, $p < .001$, 95% CI $= [0.1, 0.34]$), as participants' response times followed a flat line for the tree-broken matches ($\beta_{\text{Tree-Broken Matches}} = 0.005$, $SE = 0.05$, $p = .925$, 95% CI $= [-0.09, 0.1]$). There was also an interaction between $\log 2-$Chunk size and Target type in the tree-preserved trials ($\beta_{\text{Tree-Preserved Match} \times \text{Deviant Interaction}} = 0.12$, $SE = 0.05$, $p = .013$, 95% CI $= [0.02, 0.21]$), as participants' response times on the deviant trials were less influenced by the height of the

Table 2

Regression results in Experiment 3

MEASURE	MODEL TERM	ESTIMATE	STD. ERROR	STATISTIC	P-VALUE
Response times	Intercept	−0.02	0.15	−0.13	.897
	$\log_2$ Chunk Size	−0.22	0.03	−6.59	**<.001*** **
	Match Tree Broken	−0.09	0.11	−0.82	.415
	Deviant Tree Preserved	0.35	0.07	4.76	**<.001*** **
	Deviant Tree Broken	0.17	0.11	1.5	.13
	$\log_2$ Chunk Size × Match Tree Broken	0.22	0.06	3.69	**<.001*** **
	$\log_2$ Chunk Size × Deviant Tree Preserved	0.12	0.05	2.48	**.013***
	$\log_2$ Chunk Size × Deviant Tree Broken	0.24	0.06	3.94	**<.001*** **
Errors	Intercept	−0.1	0.12	−0.81	.421
	$\log_2$ Chunk Size	−0.25	0.05	−5.23	**<.001*** **
	Match Tree Broken	0.05	0.16	0.29	.771
	Deviant Tree Preserved	0.63	0.11	5.93	**<.001*** **
	Deviant Tree Broken	0.46	0.16	2.85	**.005** **
	$\log_2$ Chunk Size × Match Tree Broken	0.19	0.09	2.21	**.028***
	$\log_2$ Chunk Size × Deviant Tree Preserved	0.03	0.09	0.51	.609
	$\log_2$ Chunk Size × Deviant Tree Broken	0.22	0.09	2.59	**.01** **

Note. Output of the mixed model for standardized response times and error rates as a function of chunk-size ($\log_2$) and condition in Experiment 3. The tree-preserved match condition was set as the reference level. Significant p-values are written in boldface.

cut ($\beta_{\text{Tree-Preserved Deviants}} = -0.1$, $SE = 0.03$, $p = .002$, 95% CI = $[-0.17, -0.04]$). This interaction suggests that the global configurations of the tree-preserved match targets (e.g., 6 segments on the left, and 2 on the right) cannot account for the results, since on deviant trials participants did not always have to engage in tree-based operations—finding a segment in a wrong orientation sufficed to reject the target as a match.

Error rates exhibited a qualitatively similar pattern (Figure 8, bottom) but the effects and interactions were overall less pronounced. On tree-preserved match targets, participants made more errors the deeper in the tree the shapes were split ($\beta_{\text{Tree-Preserved Matches}} = -0.25$, $SE = 0.05$, $p < .001$, 95% CI = $[-0.34, -0.15]$). As predicted, the influence of chunk size was different in the tree-broken match targets ($\beta_{\text{Match Preserved} \times \text{Broken Interaction}} = 0.19$, $SE = 0.09$, $p = .028$, 95% CI = $[0.02, 0.36]$), for which chunk size played no significant role ($\beta_{\text{Tree-Broken Matches}} = -0.05$, $SE = 0.07$, $p = .428$, 95% CI = $[-0.2, 0.08]$). Unlike in the case of response times, there was no interaction with the tree-preserved deviants ($\beta_{\text{Tree-Preserved Match} \times \text{Deviant Interaction}} = 0.03$, $SE = 0.07$, $p = .609$, 95% CI = $[-0.02, 0.36]$). Here, the effect on errors was similar to that observed for the match targets ($\beta_{\text{Tree-Preserved Deviants}} = -0.21$, $SE = 0.07$, $p = .001$, 95% CI = $[-0.31, -0.12]$).

Exploratory analyses: Tree height linearity

We tested the hypothesis that the difficulty of tree-preserved matches varies as a function of tree structure with three more exploratory analyses. We conducted model comparisons to find out whether cutting deeper in the tree structure of an image gives rise to a *linear* increase in response times (corresponding to a

constant processing cost for each traversed level in the hierarchy). We compared the linear model that uses tree height as the independent variable to a model that uses chunk size (on a linear, not logarithmic, scale) and to two further models that included quadratic and cubic tree-height terms. The Bayes Information Criterion (BIC) values ranked the models in the following order: linear-tree height model: df = 4, BIC = 469.84; quadratic tree-height model: df = 5, BIC = 478.43; linear chunk-size model: df = 4, BIC = 479.17; and cubic tree-height model: df = 6, BIC = 487.2. The linear model provided the best fit to the data, as it had the lowest BIC value. An analogous model comparison on error rates also favored the linear tree-height model, although it is less clear than in the case of response times that errors should increase linearly with tree depth.

Second, we regressed the dependent variables in the tree-preserved matches against both tree height and chunk size. Recall that our main analysis lumped together two pairs of targets. Targets in which 1 segment moved were grouped with those in which 7 segments moved (both corresponding to Height 0); similarly, targets in which 2 segments moved were grouped with those in which 6 segments moved (both corresponding to Height 1). In this analysis, these target pairs were separated. Nevertheless, tree height continued to play a significant role even when chunk size was added as a predictor (RTs: $\beta_{\text{Tree Height}} = -0.21$, $SE = 0.04$, $p < .001$, 95% CI = $[-0.28, -0.13]$; Errors: $\beta_{\text{Tree Height}} = -0.25$, $SE = 0.05$, $p < .001$, 95% CI = $[-0.34, -0.16]$). Chunk size, on the other hand, had no influence on participants' performance (RTs: $\beta_{\text{Chunk Size}} = 0.01$, $SE = 0.02$, $p = .479$, 95% CI = $[-0.04, 0.02]$; Errors: $\beta_{\text{Chunk Size}} = 0.01$, $SE = 0.02$, $p = .748$, 95% CI = $[-0.03, 0.04]$).

Third, we did not find any evidence of a difference in either response times or error rates between the targets that were grouped in the main analysis (size 1 vs. 7: $M_{\text{difference-RTs}} = 21.34$, $t(35) = 1.13$, $p = .268$, 95% CI = $[-17.22, 60.01]$, Cohen's $d = 0.19$; $M_{\text{difference-errors}} = 0.03$, $t(35) = .848$, $p = .402$, 95% CI = $[-0.04, 0.11]$, Cohen's $d = 0.14$; size 2 vs. 6: $M_{\text{difference-RTs}} = 12.45$, $t(35) = 0.66$, $p = .514$, 95% CI = $[-25.91, 50.8]$, Cohen's $d = 0.11$; $M_{\text{difference-errors}} = -0.01$, $t(35) = -0.545$, $p = .589$, 95% CI = $[-0.06, 0.03]$, Cohen's $d = 0.09$).

Discussion

Experiment 3 shows that geometric shapes are easier to reconfigure, the higher up in the tree they are cut. This indicates that shapes are not represented as mere concatenations of line segments or of image chunks, but as hierarchical tree structures. This rules out a prominent alternative explanation for Experiments 1 and 2 in terms of mere chunking: The internal representation of shapes is organized into parts within parts within parts, and this organization is hierarchical and forms a nested tree. Indeed, the *linear* effect of tree height on response times neatly shows that traversing each level in a tree structure imposes a constant

processing cost.

Performance of AI Models

As an additional exploratory analysis, we also presented the tasks in Experiments 1–3 to state-of-the-art AI models. The analysis scripts are publicly available (https://osf.io/mgt25).

While not central to the main argument of the paper, these exploratory comparisons are nevertheless highly relevant given the recently reported successes of such networks in modeling human representations of geometric shapes. Both convolutional and transformer networks have been argued to accurately model multiple aspects of human vision (Eickenberg et al., 2017; Tuli et al., 2021), although this remains highly debated (Jacob et al., 2021). In particular, Campbell et al. (2024) have shown that transformer networks, and to some extent convolutional ones, can provide a good fit to the geometric-shape data in Sablé-Meyer et al. (2022). This fit also led Campbell et al. (2024) to argue that a symbolic LoT of geometric shapes need not be postulated to account for humans' behavior. The present experiments provided a renewed opportunity to evaluate whether current AI models can indeed mimic the human representation of geometric shapes.

Methods

For Experiment 1, which included animations as stimuli, we tested a state-of-the-art video model (TimeSformer: Bertasius et al., 2021). Using spatiotemporal attention, this model achieves state-of-the-art performance on multiple video classification benchmarks. For each drawing animation, we selected 8 frames to input to the model. This has been shown in previous work to be sufficient for high accuracy in video classification tasks (Bertasius et al., 2021). The frames were sampled from the drawing animations as follows. We split each animation into two parts, corresponding to the drawing generation of the two sub-shapes in the underlying structure of the overall shape (Figure 2, top). Frame 4 always depicted the first completed sub-shape in the underlying structure, and frame 8 always depicted the full shape. Frames 1–3 and Frames 5–7 were sampled at equal distances between the first frame in each part of the animation and the final frame. After the sequential exposure to the 8 frames, we extracted the network's representation (or embedding) by obtaining the numerical activations in each unit of the last hidden layer (the one preceding the output classification layer). We also obtained the last-layer embeddings for the target images by passing them as input at the same scale and in the same orientation as in our experiment. This allowed us to obtain a sample–target dissimilarity score for each pair for which we had obtained valid data from human participants. To be able to compare target embeddings to sample embeddings without averaging the sample embedding, we passed 8 samples of each target image, as if it were a still video.

In Experiments 2 and 3, we adopted the same approach. As these experiments used static stimuli

only, we tested the same networks as Campbell et al. (2024): one convolutional neural network (ResNet50: He et al., 2015) and two large transformer networks, CLIP (Radford et al., 2021) and Dinov2 (Oquab et al., 2023). As in Experiment 1, we obtained the activation vectors in the hidden layers for all the samples and targets shown to the human participants. As in Campbell et al. (2024), we only present here the analysis for the embeddings in the last hidden layer only. The full analysis, which can be consulted at https://osf.io/mgt25, shows that virtually identical results were obtained in every hidden layer of all three networks.

We used the extracted embeddings to answer two questions. First, can AI models solve the task? To assess this, we computed a dissimilarity score for each sample–target pairs by taking the Euclidean distance between the embeddings in each pair. If the networks distinguish matches from deviants, the sample–deviant dissimilarity scores should be higher than the sample–match ones. Second, do AI models show the structural effects that we found in humans? To test this, we focused on the match trials and checked whether sample–target dissimilarities differed significantly as a function of trial type in each of the three experiments. To obtain a single analogous dissimilarity score for human, we computed their average inverse efficiency score for each stimulus (response time divided by accuracy). Before the analyses, all dissimilarity scores were standardized separately for each model.

Results

Can AI models solve the task?

Figure 9 shows the overall performance of the network in Experiments 1–3, while Table 3 presents an overview of the statistical tests we performed to check whether the neural networks distinguish matches from deviants. In Experiment 1, TimeSformer's sample–deviant dissimilarity scores were significantly higher than sample–match ones. In Experiment 2, only Dino distinguished matches from deviants on the dissimilarity metric[5]. In Experiment 3, the sample-deviant dissimilarities were significantly higher than the sample–match ones in all three networks.

Do AI models show structural effects?

To find out whether the neural networks show evidence of a structured, tree-based syntactic representation, we averaged the sample–target dissimilarities by stimulus and trial type on the match-trial subsets. We then checked whether the dissimilarity scores paralleled those of human participants with repeated-measures ANOVAs by item and paired t-tests in Experiments 1 and 2 and with a linear model

[5] ResNet and Clip also distinguished between matches and deviants significantly, but in earlier layers: https://osf.io/mgt25.

Table 3

AI Models: Statistical tests on distinguishing match versus deviants

EXPERIMENT	AI MODEL	$M_{\text{DIFFERENCE}}$	t-VALUE	DF	p-VALUE	95% CI	EFFECT SIZE
1	TimeSformer	0.28	6.02	15	**< .001*** **	[0.18, 0.38]	1.5
2	ResNet	0.03	0.65	31	.517	[−0.06, 0.11]	0.12
2	Clip	0.02	0.14	31	.888	[−0.25, 0.29]	0.03
2	Dino	0.23	2.54	31	**.016***	[0.05, 0.42]	0.45
3	ResNet	0.26	3.1	15	**.007****	[0.08, 0.44]	0.78
3	Clip	0.38	3.93	15	**.001****	[0.17, 0.58]	0.98
3	Dino	0.37	3.59	15	**.003****	[0.15, 0.59]	0.9

Note. Results of t-tests on the networks' paired dissimilarity scores (deviants − matches) by experiment and AI model. Significant p-values are written in boldface.

with random item intercepts in Experiment 3. Evidence of syntactic structure in the networks' representations would be obtained if the dissimilarity scores between sample and targets were higher for incongruent than for congruent trials in Experiment 1, higher for Different-subtrees than Same-subtree in Experiment 2, and inversely related to Tree Height in Experiment 3.

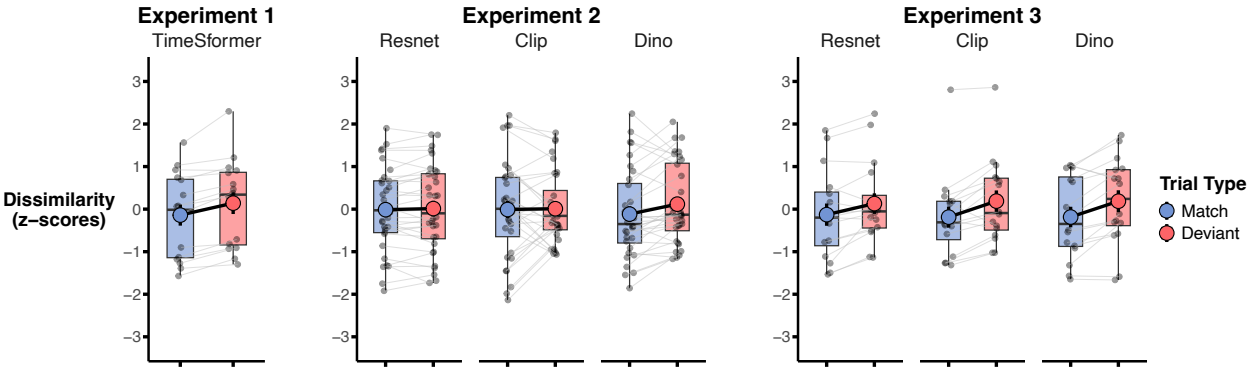
We found no evidence of syntactically structured representations in any of the networks across the three experiments (Figure 10). Experiment 1 revealed a Model × Trial type interaction, $F(1, 3) = 19.38$, $p < .001$, $\eta_p^2 = .56$. This interaction occurred because humans' dissimilarity scores were significantly different for congruent versus incongruent trials, $M_{\text{difference}} = 0.76$, $t(15) = 4.36$, $p < .001$, 95% CI $[0.39, 1.13]$, Cohen's $d = 1.09$, whereas no such difference emerged for TimeSformer, $M_{\text{difference}} = 0.003$, $t(15) = 0.14$, $p = .891$, 95% CI $[-0.04, 0.04]$, Cohen's $d = 0.03$. This lack of a difference was observed in all of TimeSformer's hidden layers.

Similarly, in Experiment 2, dissimilarity scores exhibited a Model × Trial type interaction, $F(3, 93) = 4.73$, $p = .004$, $\eta_p^2 = .13$. Again, the difference between congruent and incongruent trials was significant in the human dataset, $M_{\text{difference}} = 0.81$, $t(31) = 3.42$, $p = .002$, 95% CI $[0.33, 1.29]$, Cohen's $d = 0.61$, but inexistent in any of the AI networks (ResNet: $M_{\text{difference}} = 0.05$, $t(31) = 0.19$, $p = .852$, 95% CI $[-0.35, 0.45]$, Cohen's $d = 0.05$; Clip: $M_{\text{difference}} = 0.02$, $t(31) = 0.22$, $p = .827$, 95% CI $[-0.19, 0.24]$, Cohen's $d = 0.04$; Dino: $M_{\text{difference}} = 0.09$, $t(31) = 0.7$, $p = .491$, 95% CI $[-0.18, 0.36]$, Cohen's $d = 0.12$). This pattern of (null) results held for all hidden layers of all the three networks.

Finally, in Experiment 3, the influence of $\log_2$-chunk size (tree height) on the match targets with preserved tree structure was significantly below 0 for humans, $\beta_{\text{Tree-Preserved Matches}} = -0.37$, $SE = 0.08$,$p < .001$, 95% CI $= [-0.52, -0.22]$. There were pairwise interactions on the influence of $\log_2$-chunk size between the human participants and each of the three networks (Humans × ResNet:

Figure 9

AI models generally distinguish matches from deviants in all three experiments



Note. Sample–Target dissimilarity scores by target type in Experiments 1–3. Gray circles and the lines connecting them represent averages by stimulus, as a function of target type; colored circles depict group averages ±1 SEM; boxplots indicate the median and interquartile range.

$\beta_{\text{Tree-Preserved Matches}} = 0.39$, $SE = 0.11$,$p < .001$, 95% CI = $[0.18, 0.6]$; Humans × Clip: $\beta_{\text{Tree-Preserved Matches}} = 0.85$, $SE = 0.11$, $p < .001$, 95% CI = $[0.64, 1.06]$; Humans × Dino: $\beta_{\text{Tree-Preserved Matches}} = 0.79$, $SE = 0.11$, $p < .001$, 95% CI = $[0.58, 1]$). The $\log_2$-chunk size coefficients for each AI model indicate that tree height played no role in the convolutional network (ResNet: $\beta_{\text{Tree-Preserved Matches}} = 0.02$, $SE = 0.08$,$p = .83$, 95% CI = $[-0.13, 0.17]$) but correlated *positively* with distance in the transformer networks (Clip: $\beta_{\text{Tree-Preserved Matches}} = 0.48$, $SE = 0.08$, $p < .001$, 95% CI = $[0.33, 0.63]$; Dino: $\beta_{\text{Tree-Preserved Matches}} = 0.42$, $SE = 0.08$, $p < .001$, 95% CI = $[0.27, 0.57]$). All hidden layers of the three networks exhibited this pattern of results.

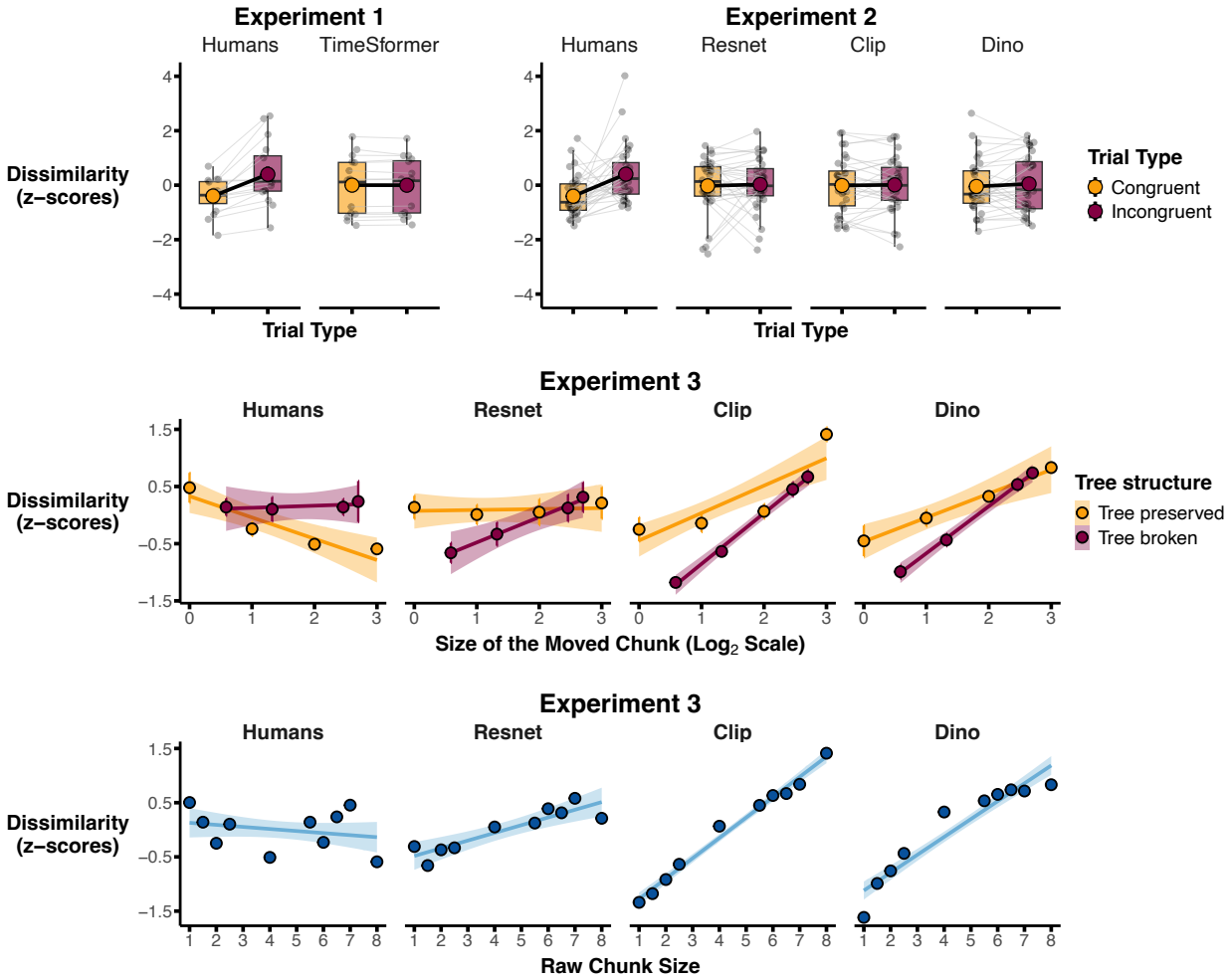
Those findings suggests that transformer networks prioritize pixel overlap over other criteria in their internal representations: For them, unlike humans, the similarity between samples and targets is higher when fewer segments are moved from the sample to the target (Figure 10, bottom row). Indeed, when regressing dissimilarity scores against the raw size of the moved chunk, we found that this metric did not predict human performance ($\beta_{\text{Raw chunk size}} = -0.04$, $SE = 0.02$, $p = .089$, 95% CI = $[-0.08, 0.01]$) but was strongly and positively related to the dissimilarity scores of all three networks (ResNet: $\beta_{\text{Raw chunk size}} = 0.14$, $SE = 0.02$,$p < .001$, 95% CI = $[0.1, 0.19]$; Clip: $\beta_{\text{Raw chunk size}} = 0.38$, $SE = 0.02$, $p < .001$, 95% CI = $[0.33, 0.42]$; Dino: $\beta_{\text{Raw chunk size}} = 0.33$, $SE = 0.02$, $p < .001$, 95% CI = $[0.29, 0.37]$).

Discussion

Neither convolutional nor transformer networks showed any evidence of hierarchical representations, even though the networks distinguished matches from deviants. These results suggest that current AI architectures do not build human-like representations of visual shapes, especially for out-of-sample stimuli

Figure 10

AI Models do not show any evidence of a tree-based structural representation of shapes



Note. Sample–Match dissimilarity scores, averaged by stimulus, by Trial type, Model (Human or AI network), and Experiment. Top row: Experiments 1 and 2. Middle row: Experiment 3 with dissimilarity scores plotted as in the main analysis (tree height for tree-preserved matches; the logarithm of the moved-chunk size for tree-broken matches). Bottom row: Same data but plotted as a function of the moved chunk size. AI model similarity is determined by the size of the extract part, whereas human behavior is driven by whether and at which height a part appears in a shape's tree structure.

such as the ones used here. We do not wish to overemphasize this claim, notably because many of the geometric shapes in the present study were new to the AI networks we tested. Differences between humans and artificial neural networks might decrease in upcoming models, especially if the stimuli and tasks of Experiments 1–3 become part of future training sets.

If that should happen, however, it would not necessarily undermine the language of thought hypothesis. While such an outcome might indicate that the human data can be accounted for without resorting to the LoT hypothesis, which is the path taken by Campbell et al. (2024), this outcome could also mean that the networks implement or approximate a language of thought (Wu et al., 2024). For instance, large language models (LLMs) have been found to implicitly compute syntactic structures consistent with linguistic theories (Diego-Simón et al., 2024; Hewitt & Manning, 2019). Thus, the absence of explicit symbolic operations in neural networks does not preclude the possibility that they compute over internal representations that functionally resemble a LoT. Positive findings, taken by themselves, are thus more ambivalent than negative findings, such as the ones reported here. Overall, the present work suggests that the human cognitive architecture seems, for the moment at least, to be uniquely predisposed to internally represent a broad class of stimuli as language-like tree structures (Dehaene et al., 2015; Fitch, 2014; Hauser & Watumull, 2017).

General Discussion

Overview

By itself, the hypothesis of hierarchical structure in visual perception is not new (e.g., Biederman, 1987; Cavanagh, 2021; Palmer, 1977). Almost 50 years ago already, seminal work by Palmer (1977) indicated that the relation between the segments of a contour shape might be represented in a hierarchical tree structure.

Our study complements this line of work in several ways. First, we bridge geometric shape perception to work on the Language of Thought hypothesis, which has relied primarily on stimulus complexity as a proxy for internal representation. We provide more direct evidence for LoT representations in the domain of geometry through constituency tests that closely mirror those that linguists have been using to probe the syntactic structure of natural-language representations: structural ambiguity (Experiment 1: the same shape can be assigned different structural representations), subtree facilitation (Experiment 2: shape subparts are easier to recognize when they belong to the same subtree), and movement at various syntactic depths (Experiment 3: shape fragments are easier to reconfigure the higher in the tree they are split). This way of testing for syntactic structure is not unique to geometric shapes, and we hope to see such tests extended to other cognitive domains as well. In each experiment, we took great care to design the stimuli in a way that rigorously controls for alternative explanations. Moreover,

the converging findings from three distinct paradigms provide strong evidence that the representations adults use for geometric shapes have syntactic structure.

At the same time, the evidence favoring syntax casts doubt on competing alternatives. Experiments 1 and 2 rule out the possibility that geometric-shape representations are unstructured—otherwise, the representations assigned to one shape would not have varied in Experiment 1, and there would not have been any facilitation effect for same-subtree subparts in Experiment 2. Experiment 3 also rules out the chunk-concatenation account by demonstrating that shapes are represented as hierarchical tree structures—subunits at each level of the hierarchy are themselves formed of nested subunits, all the way down to individual line segments, as predicted by the LoT proposed by Sablé-Meyer et al. (2022).

Another advantage of our paradigms is that they tapped into mental representations implicitly, without asking participants to make explicit judgments about the structure of visual shapes. By contrast, most of the experiments reported in Palmer (1977) asked for explicit subjective judgments, e.g., about the subjective parsing of a shape (E1), about the subjective "goodness" of its parts (E2), or about the subjective completion of a mental image (E4). Only his Experiment 3 was very similar to our Experiment 2, since it recorded accuracy and response times for the verification of whether a subpart belongs to a shape. But even in this case, the range of shapes we tested was much broader than the grid-world shapes in Palmer (1977), and our design fully counterbalanced the shapes and their subparts: Every sample shape and every target subpart appeared equally often on congruent and incongruent trials, and on match and deviant trials, respectively. Considering all experiments in our study together, we go beyond Palmer (1977) by testing ambiguous shapes in the context of the shape LoT in Experiment 1 and by investigating multiple levels of shape hierarchy in Experiments 2 and 3.

Another source of innovation comes from the availability of a specific LoT for a broad range of geometric shapes (Sablé-Meyer et al., 2022), which our findings strengthen the evidence for. In particular, the decompositions we hypothesized for the stimuli in Experiment 2 follow from efficiency considerations that derive from the particular LoT implementation in Sablé-Meyer et al. (2022).

Our third contribution consists in making progress on the debate concerning perception and representation in artificial neural networks (Campbell et al., 2024; Sablé-Meyer et al., 2021). The embeddings of convolutional and transformer models revealed that deep networks could perform above chance level in our tasks—distinguishing matches from deviants—yet failed to exhibit any evidence of syntactic encoding. This finding contrasts with recent work using the same measures and models and claiming that deep networks could account for aspects of human behavior in geometry perception, despite the lack of an internal symbolic language (Campbell et al., 2024). As this AI–human alignment stemmed

from prior work focusing on complexity measures, the findings suggest that AI representations may be sensitive to minimum description length through compression mechanisms that do not involve syntactic representations. This reinforces our point that complexity measures are indirect and limited in the amount of information that they convey about the structure of internal representations. In future research, we hope to see constituency tests of the kind used here extended to other cognitive domains that have been argued to be driven by LoT representations (e.g., Piantadosi et al., 2016).

Open questions

The current results thus place the language of thought hypothesis for geometric shapes on solid footing. However, they are only the first step toward a full theory of geometric shape representations. The evidence we report here belongs to the *computational* level of analysis (Marr, 1982), as it only addresses the relation between the input (geometric shapes) and the output representations (syntactic trees). While we show that shape representations are syntactic, the experiments are silent about the *algorithm* that turns retinal input into tree-structured LoT representations. So far, Bayesian program induction or generative models that infer the underlying structure of stimuli are the best candidate for such inferences (Ellis et al., 2023; Froyen et al., 2015; Lake et al., 2015), especially considering the nonsymbolic AI models' failures in our tasks. We leave it to future research to bridge the algorithmic gap between the visual inputs and the tree-structured outputs we found evidence of. Needless to say, our behavioral experiments are also silent on questions concerning neural *implementation*. Although previous work found both neural correlates of LoT complexity and potentially dedicated cerebral areas for LoT representations (e.g., Al Roumi et al., 2023; Sablé-Meyer et al., 2025), how symbolic representations in general, and geometric shape representations in particular, are implemented in the human brain remains to be understood.

Another question that our study leaves open concerns the dependency of the LoT representations on the visual modality. While the LoT representations that we postulate have been proposed as a language for *visual* shapes, this language comprises abstract *amodal* instructions (e.g., REPEAT, CONCATENATE) that might extend beyond the visual domain. Indeed, these instructions can not only encode visual shapes, but also redraw them via commands sent to the motor system. In this respect, our proposal is similar to Lake et al. (2015) and Sablé-Meyer et al. (2022) in assuming that perceiving a geometric shape is a form of program inference—that is, of identifying a program, possibly the shortest, that could have drawn the shape. If this is correct, then the format of representation of geometric shapes might well be nonspecific to the visual modality. This possibility is supported by the fact that congenitally blind subjects have comparable geometric intuitions to sighted subjects (Amalric et al., 2018) and exhibit a similar geometric complexity effect as sighted subjects when performing a quadrilateral intruder test in the tactile modality

(Adriano et al., 2025).

Furthermore, while convolutional neural networks provide a reasonable model of early vision (Yamins & DiCarlo, 2016), their failure to emulate the human data in the present experiments might be used as preliminary evidence against a purely visual account. However, the extent of the similarities between human minds and deep networks remains contentious (Xu & Vaziri-Pashkam, 2021), so further experiments are required to address the locus and modality specificity of the geometry LoT.

Obtaining analogous effects in a different modality (e.g., tactile) or locating the source of the effects in a nonvisual brain area would provide support for an amodal locus of geometry LoT representations. Ongoing work in our laboratory points in this direction: The same LoT postulated by Amalric et al. (2017) for geometric spatial sequences was found to readily explain the perception of auditory two-tone sequences (Al Roumi et al., 2023), and fMRI and MEG recordings suggest that, beyond the visual cortex proper, such sequences recruit neuronal circuits of the bilateral intraparietal sulcus and posterior inferior temporal gyrus that overlap with those involved in number and mathematics (Al Roumi et al., 2023; Sablé-Meyer et al., 2025). Obtaining a firm answer to this question would be relevant for the long-standing debate on the iconic versus symbolic nature of perceptual representations (Kosslyn, 1981; Pylyshyn, 1981) and on the controversial divide between perception and cognition (Block, 2023; Quilty-Dunn et al., 2023).

Constraints on generality

One last limit of the present work is that we do not know whether the current results would generalize beyond the English-speaking, educated and digitally literate adults in our sample, which we tested due to ease of access. The present experimental designs provide a framework for evaluating geometric shape representations in other adult populations, as well as in infants, children, and non-human animals. These extensions are needed to gain a more complete picture of the scope, development, dependence on education, and human uniqueness of LoT representations for geometry.

Relation to other theories on shape representation

Our results open new avenues for thinking about the status of the Gestalt principles that underlie perceptual organization, such as closure, parallelism, and good continuation (Wertheimer, 1938). These principles have been rightly criticized for being vague and for providing redescriptions rather than explanations (for a review, see Wagemans et al., 2012). The geometry LoT offers a potential explanation by providing a formalization of how such individual features get assembled into higher-level structures, from which some Gestalt principles follow. This applies not only to Gestalt theories of perception but also to more recent work arguing that object individuation follows from the syntactic structure of visual scene representations (Feldman, 2003). At present, however, such proposals (including the present one) typically

rely on experiments that use geometric shapes, not objects, as stimuli (e.g., Feldman, 2007). Thus, the generalizing leap from geometric shapes to ordinary objects and scenes will require more research.

Another open question concerns the interaction between LoT representations and other shape representations. In agreement with a recent "pluralistic" view of visual representations (Green, 2023), we find it likely that the geometry LoT is only one of the many levels of visual representations. Here, in order to force participants to focus on high-level representations, and prevent their reliance on early shape representations in iconic or visual short-term memory (Luck & Vogel, 1997; Sperling, 1960), we used transformations such as scaling, rotation, and translation (Experiments 1 and 2) and a post-sample visual mask (Experiment 3). Thus, low-level visual representations of shapes were excluded by design.

Higher-level structural representations different from the geometry LoT we have tested may have been evoked, however. For instance, shape-skeleton models (Blum, 1967; Feldman & Singh, 2006) argue that shapes are represented via a set of internal axial branches. In the deterministic version of these models (Blum, 1967), these branches are the shape's medial axes—the set of internal points with more than one closest point on the shape's boundary. In the Bayesian framework of the model (Feldman & Singh, 2006), these branches are determined by a combination of the shape's medial axes and prior knowledge about likely skeletons, which favors simpler structures. Skeletons are typically accompanied by metric information, but at the most abstract level, only their topology may be encoded (Yousif & Brannon, 2024, 2025): how many branches they possess and how the branches connect (e.g., number of T-junctures, number of L-junctures, etc.).

However, neither shape-skeleton nor topological models seem to be able to account for our results. In Experiment 1, the shapes did not vary in the number or type of segment junctures across conditions, but only in the preceding drawing process—a dynamic factor that neither of these alternative models considers. By contrast, the geometry LoT is implicitly dynamic, as its primitives correspond to motor drawing commands (e.g., TRACE a segment, TURN the drawing angle). Furthermore, in Experiments 2 and 3, all the shapes were made of line segments assembled in branches. According to the deterministic shape skeleton model (Blum, 1967), which does not incorporate prior probabilities, these branches should coincide with the shape's medial axis and thus with its skeleton. To explain our results, one would have to assume that this skeleton is itself represented as a syntactically structured representation, as in the LoT model, an extension that is currently missing from the shape skeleton model. In the probabilistic shape skeleton model (Feldman & Singh, 2006), the shapes' medial axis need not coincide with their shape skeletons. However, as the stimuli in Experiment 2 were designed in matched pairs that perfectly overlapped in segment subsets, it is hard to see why skeletal representations would deviate radically from each other, given that they originate in the same local visual input. This argument applies to the topological model as well, as

the number and type of junctures were identical in the same-subtree/different-subtree contrasts we tested.

Finally, for Experiment 3, it is unclear what predictions the shape skeleton and topological models would make. Previous work seems to indicate that changes to the central node of a skeleton makes the detection of a difference easier for adults (e.g., Lowet et al., 2018). We did not have an equivalent test in Experiment 3, where we cut the shape instead of altering it. However, the fact that undoing a cut at the central node (Height 1) was easier than undoing a cut at an arm (Height 0) provides tentative evidence against the skeleton account. Similarly, we did not design Experiment 3 to contrast the geometry LoT account with the topological-relations model (Yousif & Brannon, 2025), which also makes a direct comparison challenging. Note, however, that while in Experiment 3, the tree-broken matches preserved the *number* of segment junctures, adults took longer to identify them as matches than the tree-preserved ones, which changed the number of junctures.

These observations suggest that existing shape skeleton and topological models do not readily account for our results, but they do not imply that these models are wrong, especially as the present experiments were not designed to tease them apart. On the one hand, different stimuli classes might give rise to different representations: LoT representations for regular geometric shapes, shape-skeleton representations for biological and object shapes, and topological relations for map-like structures. On the other, shape skeletons and topological relations may first be extracted by independent means, then compressed using the geometry LoT for more efficient storage (as proposed by Sablé-Meyer et al., 2022). We think that the latter proposal has much in its favor: Shape skeletons may be an efficient representation of shapes, particularly biological ones, but they can be further compressed still. This fusion between the shape skeleton and the geometry LoT proposals might account for considerably more data than either proposal alone. Moreover, the Bayesian shape skeleton model of Feldman and Singh (2006), while not conceived as an LoT , incorporates a nested hierarchy of roots, branches, and sub-branches, unlike its deterministic predecessor (Blum, 1967). This hierarchical constituent structure, inherent in the generative nature of the algorithm that produces skeletons, partly aligns the shape skeleton model with the geometry LoT tested here. Indeed, it has recently been argued that shape skeleton representations (Green, 2023; Hafri et al., 2023), as well as other high-level object representations (e.g., Barenholtz & Tarr, 2008), exhibit signature LoT features (compositionality, productivity, systematicity) and should thus be thought of as such (Quilty-Dunn et al., 2023). If true, these various representations should be relatively straightforward to integrate.

At present, however, existing skeletal or topological models do not offer a straightforward explanation for why a zigzag looks more regular to humans than a random arrangement of the same sequence of line segments. We propose that the geometry language may have emerged, perhaps uniquely in

humans, as an additional layer that compresses animal, plant, and other natural skeletons according to their numerical regularities and symmetries. Together with a human-specific capacity for understanding visual objects as communicative symbols (Deacon, 1997; Revencu, 2024), this would explain why only humans create drawings of skeleton shapes, and why these drawings tend to be more geometrically regular than their real-world referents (e.g., depicting a river as a regular zigzag, or a quadruped as a horizontal segment with four symmetrical, perpendicular legs).

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